

IDENTIFICATION

PRODUCT CODE: DEC-8E-EUZC-D
PRODUCT NAME: TDS-E DECTAPE FORMATTER
DATE CREATED: DECEMBER 7, 1971
MAINTAINER: DIAGNOSTIC PROGRAMMING GROUP
AUTHOR: BRUCE HANSEN

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1. ABSTRACT

THE TDB-E DECTAPE FORMATTER PROGRAM RECORDS THE TIMING AND MARK TRACKS ON A DECTAPE MOUNTED ON THE TU56 DECTAPE TRANSPORT.

THE PROGRAM INTERACTS WITH THE OPERATOR VIA THE TELETYPE TO OBTAIN THE NECESSARY DATA FOR EACH SET OF DECTAPES TO BE FORMATTED. AS SOON AS ONE SET OF TAPES IS FORMATTED, THE PROGRAM IS READY TO FORMAT ANOTHER SET.

THREE FULL PASSES ARE REQUIRED TO COMPLETELY FORMAT EACH DECTAPE, AND UP TO TWO DECTAPES MAY BE FORMATTED AT A TIME (UNITS 0 AND 1 WITH A TDB-E; IOT CODE OF 677X). UPON COMPLETION OF A CYCLE, NEW TAPES MAY BE MOUNTED AND FORMATTED AS THE LAST, WITH A MINIMUM OF OPERATOR-PROGRAM COMMUNICATION. ONE TAPE EXCLUDING TAPE SETUP TIME, REQUIRES THREE MINUTES FROM START TO FINISH.

2. REQUIREMENTS

2.1 EQUIPMENT

PDP-8E, TELETYPE, TDB-E (IOT CODE 677X), AND A TU56 DECTAPE TRANSPORT.

2.2 STORAGE

THIS PROGRAM USES LOCATIONS 0000-3400. THE LOADERS MUST BE STORED IN THE LAST MEMORY PAGE.

2.3 PRELIMINARY PROGRAMS

ALL BASIC PDP-8E DIAGNOSTIC PROGRAMS AND MAINDEC-8E-03A(N) SHOULD HAVE BEEN SUCCESSFULLY RUN.

3. LOADING PROCEDURE

LOAD THE PROGRAM INTO FIELD 0 USING THE STANDARD BINARY LOADER.

4. STARTING PROCEDURE

4.1 STARTING ADDRESS

SET SWITCH REGISTER TO 0200 AND PRESS LOAD ADDRESS. NOW PRESS CLEAR AND THEN CONTINUE. "UNIT" IS PRINTED ON THE TELETYPE.

MOUNT THE DECTAPES TO BE MARKED ONTO THE TAPE TRANSPORTS. WITH JUST ENOUGH TURNS OF TAPE ON THE RIGHT HAND REEL OF EACH TRANSPORT TO PROVIDE A GRIP. MAKE SURE THAT NO TWO TAPE UNITS ARE SET TO THE SAME UNIT NUMBER. SET THE SWITCH ON THE TDB-E TO WTH POSITION. FOR EACH TRANSPORT TO BE USED, SET THE WRITE ENABLED-WRITE LOCK SWITCH TO WRITE ENABLED, AND THE REMOTE-OFF-LOCAL SWITCH TO REMOTE.

OPERATING PROCEDURE

THE PROGRAM AND OPERATOR NOW CONVERSE. THE PRINTOUT "UNIT?" IS ASKING WHICH DECTAPE UNITS WILL BE USED. THE OPERATOR TYPES ONE OR TWO UNIT NUMBERS, CORRESPONDING TO THE DECTAPE UNITS UPON WHICH HE HAS MOUNTED TAPES. FOR INSTANCE, IF THE OPERATOR HAS MOUNTED TAPES ON UNITS 0 AND 1, HE WOULD TYPE 0 1- (WHERE - SIGNIFIES CARRIAGE RETURN). SPACES ARE IGNORED, SO IT MAKES NO DIFFERENCE IF THE OPERATOR TYPES SPACES BETWEEN THE UNIT NUMBERS. ONLY ONE SPECIFICATION OF A UNIT IS SIGNIFICANT, I.E., TYPING 000111 HAS THE SAME EFFECT AS TYPING 01.

ONCE THE OPERATOR HAS SPECIFIED THE UNITS HE WISHES TO USE, THE PROGRAM TYPES "FORMAT?". THE OPERATOR RESPONDS BY TYPING MARK OR MARK XXXX-. IF HE TYPES MARK-, THE PROGRAM ASSUMES 201 WORDS 2702 BLOCKS (STANDARD PDP-8 FORMAT). OTHERWISE XXXX IS ACCEPTED AS A DECIMAL NUMBER OF WORDS PER BLOCK AND MUST BE DIVISIBLE BY 3. NOTE THAT TYPING MARK 384 WILL CAUSE THE PROGRAM TO GENERATE A STANDARD PDP-10 FORMAT DECTAPES (1102(8) BLOCKS OF 600 WORDS, WHICH IS EQUIVALENT TO 1102(8) BLOCKS OF 200 WORDS WHERE EACH WORD IS 36 BITS RATHER THAN 12 BITS).

THE PROGRAM NOW TYPES "XXXX WORDS, YYYY BLOCKS OK? (YES OR NO)". THIS SERVES AS A FINAL CHECK FOR BLOCK COUNT. XXXX AND YYYY ARE OCTAL VALUES REPRESENTING THE FINAL OUTCOME OF A FORMULA SOLVED BY THE PROGRAM, DETERMINING THE NUMBER OF BLOCKS THAT MAY BE WRITTEN ON A DECTAPE KNOWING THE NUMBER OF WORDS. IF A NO ANSWER IS GIVEN, THE PROGRAM REVERTS TO "FORMAT?". OTHERWISE (IF YES), THE PROGRAM TYPES OUT "SET SWITCH TO WTM". THEN THE OPERATOR HITS CARRIAGE RETURN ON THE TELETYPE AND THE TAPE ON FIRST UNIT SPECIFIED BEGINS TO MOVE IF THE SWITCH IS SET.

ONCE ALL OF THE TAPES SPECIFIED HAVE BEEN MARKED, THE PRINTOUT "SET SWITCH TO OFF" APPEARS. THEN THE OPERATOR RESETS THE WTM SWITCH TO OFF, AND STRIKES THE RETURN KEY ON THE TELETYPE, STARTING THE SECOND PASS. NOTE THAT DURING THE SECOND PASS WITH MULTIPLE DECTAPE UNITS, AS SOON AS ONE TAPE STOPS AND THE NEXT TAPE STARTS, THE FIRST TAPE IS COMPLETED AND MAY BE REPLACED WITH A FRESH TAPE IN PREPARATION FOR RECYCLING.

THE PROGRAM CONTINUES BY ITSELF UNTIL COMPLETED, AT WHICH TIME THE "FORMAT" PRINTOUT OCCURS. TYPING "SAME-" REPEATS THE ENTIRE PROCESS WITH THE ORIGINAL CONSTANTS. THE NEW DECTAPES MUST BE MOUNTED AND READY TO WRITE TIMING AND MARK TRACKS BEFORE A CARRIAGE RETURN IS HIT ON THE TELETYPE AFTER THE TYPEOUT "SET SWITCH TO WTM". ALSO, IN RESPONSE TO "DIRECT?", TYPING "RDR-" CAUSES THE PRINTOUT OF THE UNIT NUMBER OF THE DECTAPE AND THE LAST 22 BLOCK NUMBERS; "RDF-" CAUSES THE PRINTOUT OF THE UNIT NUMBER AND THE FIRST 22 BLOCK NUMBERS; AND "RESTART-" RETURNS THE PROGRAM TO "UNIT?" UNIT NUMBERS ARE PRINTED AS "000N" WHERE N IS THE UNIT NUMBER.

FOLLOWING ARE SEVERAL EXAMPLES OF SUCCESSFUL OPERATION. THE UNDERLINED STATEMENTS ARE PRINTED BY THE PROGRAM. ALL OPERATOR RESPONSES SHOULD BE FOLLOWED BY A CARRIAGE RETURN.

- A. CREATE A STANDARD PDP-8 TAPE ON UNIT 1
UNIT? 1
FORMAT? MARK
0201 WORDS, 2702 BLOCKS, OK? (YES OR NO)

YES
SET SWITCH TO WTM
SET SWITCH TO OFF
FORMAT?

B. CREATE 4 STANDARD POP-10 FORMAT TAPES, TWO AT A TIME ON UNITS 0-1

UNIT? 01
FORMAT? MARK 384
0600 WORDS, 1102 BLOCKS OK? (YES OR NO)
 YES
SET SWITCH TO WTM
SET SWITCH TO OFF
FORMAT? SAME
SET SWITCH TO WTM
SET SWITCH TO OFF
FORMAT?

4.3 ERRORS

4.3.1 ERRORS TYPED TO "UNIT" AND "FORMAT" REVERT BACK TO "UNIT?" OR "FORMAT?"

4.3.2 ERROR MESSAGES FOR RESPONSE TO MARK XXXX

NOT DECIMAL	A CHARACTER IN XXXX IS NOT 0-9
NOT DIVISIBLE BY 3	XXXX CANNOT BE DIVIDED EVENLY BY 3
TOO MANY WORDS	THE NUMBER OF WORDS PLUS 15 EXCEEDS 7777(8).
TOO MANY BLOCKS	THE NUMBER OF BLOCKS GENERATED BY XXXX EXCEEDS 7777

4.3.3 ERROR MESSAGES FOR RESPONSE TO "SET SWITCH TO WTM".

1. SETUP? INDICATES AN ERROR IN THE DECTAPE SETUP (SEE SECTION 4.1 FOR DECTAPE SETUP)
 ONE OF THE UNITS SPECIFIED IS IN WRITE LOCK POSITION, NOT SELECTED,
 OR THE WRITE FLIP-FLOP IS UNABLE TO BE SET, OR THERE MAY BE A TIMING ERROR.
 (AFTER MESSAGE REVERT BACK TO "UNIT")
2. SWITCH NOT SET TO WTM OR SINGLE LINE FLAG FAILED TO SET
 SET SWITCH TO WTM.

THIS TYPE OUT SAYS THAT EITHER THE SWITCH ON THE M868 MODULE IS NOT SET TO THE WTM POSITION OR THE TIMING GENERATOR FOR WRITING THE MARK AND TIMING TRACKS IS NOT SETTING THE SINGLE LINE FLAG.

RECOVERY:

IF THE SWITCH WAS NOT SET TO WTM POSITION SET THE SWITCH AND HIT CARRIAGE RETURN ON THE TELETYPE.

IF THE SWITCH WAS SET TO WTM POSITION AND THIS TYPE OUT OCCURRED, TRY AGAIN OR EXAMINE THE TIMING GENERATOR CIRCUIT.

4.3.4 ERROR MESSAGES FOR MARKING AND VERIFYING A TAPE

PC XXXX MARK TRACK ERROR PHASE Y
PC XXXX BLOCK NUMBER ERROR PHASE Y
PC XXXX DATA ERROR PHASE Y
PC XXXX CHECKSUM ERROR PHASE Y
PC XXXX TIMING ERROR PHASE Y
PC XXXX WRITE ERROR PHASE Y

XXXX EQUALS THE PROGRAM COUNTER AT TIME OF THE FAILURE.
Y EQUALS THE PASS WHICH IT WAS IN. (SEE SECTION 4.4)
RECOVERY

4.4

ALTHOUGH AN ERROR SHOULD CAUSE DOUBT CONCERNING THE ENTIRE PROCESS,
A RESTART MAY BE MADE (EXCEPT IN PHASE 0) BY TYPING "RETRY."
RETRY CAUSES THE PROGRAM TO GO BACK TO PHASE 1, TYPE "RESTART" TO RETURN TO "UNIT?"

PHASE 0: WRITE TIMING AND MARK TRACK FORWARD
PHASE 1: READS MARK TRACK REVERSE
PHASE 2: WRITE DATA, FORWARD BLOCK AND REVERSE BLOCK NUMBERS FORWARD AND WRITES THE CHECKSUMS
PHASE 3: DISPLAYS BLOCK NUMBERS IN AC REVERSE
PHASE 4: READS DATA, FORWARD BLOCK AND REVERSE BLOCK NUMBERS FORWARD AND CALCULATES THE CHECKSUM
PHASE 5: READS REVERSE BLOCK NUMBERS IN REVERSE

THE ENTIRE PROGRAM MAY BE RESTARTED AT 0200 ANY TIME.

5.

DETAILS OF OPERATION AND STORAGE

THE PROGRAM WRITES TIMING AND MARK TRACK ON A DECTAPE FORWARD WITH WITH SWITCH SET, THEN IT READS THE MARK TRACK IN THE REVERSE DIRECTION WITH THE SWITCH SET TO OFF. THE PROGRAM CHECKS ALL OF THE MARK TRACK ONCE IT IS IN SYNC. (SEE FLOW FIGURE 1) WHEN IT FINISHES READING THE MARK TRACK REVERSE, IT BOUNCES OFF THE END ZONE AND STARTS WRITING ZEROS TO THE FIRST BLOCK MARK. THE PROGRAM IS NOW IN SYNC. THE PROGRAM NOW CONTINUES WRITING FORWARD BLOCK NUMBERS, REVERSE CHECKSUM, DATA, CHECKSUM, AND REVERSE BLOCK NUMBERS FOR THE REST OF TAPE. WHEN IT SEES THE END ZONE, IT TURNS AROUND AND STARTS DISPLAYING THE REVERSE BLOCK NUMBER IN THE ACCUMULATOR UNTIL IT HITS THE END ZONE AGAIN. NOW THE TAPE TURNS AROUND AND STARTS READING AND COMPARING ALL FORWARD BLOCK NUMBERS, REVERSE CHECKSUM, ALL DATA, CHECKSUM AND REVERSE BLOCK NUMBERS THAT WAS WRITTEN IN PHASE 2. THIS COMPARISON IS DONE ON ALL BLOCKS UNTIL THE END ZONE IS REACHED. THE TAPE TURNS AROUND IN THE END ZONE AND STARTS LOOKING FOR REVERSE BLOCK NUMBERS AND COMPARING THEM ALL THE WAY DOWN TAPE TO THE END ZONE. THE FORMATTING IS NOW COMPLETE, THE TAPE STOPS, AND "FORMAT" IS TYPED OUT WAITING FOR NEW DIRECTIONS.

THE NUMBER OF BLOCK FRAMES TO BE WRITTEN IS A FUNCTION OF THE NUMBER OF WORDS PER BLOCK
THE FORMULA

$$\text{BLOCKS PER TAPE} = \left\lceil \frac{(212080)/(NW+15)}{2} \right\rceil$$

WHERE NW EQUALS THE NUMBER OF WORDS TO BE WRITTEN, IS USED BY THE PROGRAM TO COMPUTE THE NUMBER OF BLOCKS, BUT IS ADJUSTED BY THE PROGRAM TO PROVIDE THE STANDARD PDP-8 FORMAT OF 129(10) (12-BIT) WORDS, 1474(10) BLOCKS, AND STANDARD PDP-10 FORMAT OF 128(10) (36-BIT) WORDS, 578(10) BLOCKS.

5.1

THEORY

THE WRITING OF THE MARK TRACK IS DONE THROUGH AC BITS 0, 3, 6 AND 9, THE FOLLOWING DESCRIPTION IS HOW THE MARK TRACK IS WRITTEN.

- A. INSTALL THE TAPE WITH ENOUGH TURNS TO CREATE A PULL. THE REVERSE END ZONE REQUIRES A SEQUENCE OF THREE DATA WORDS FOR ITS PATTERN.

4044
0440
4404

IN THE MARK TRACK THE WORDS APPEAR AS 101101101101 (5555(8)). THE REVERSE END ZONE SHOULD COVER ABOUT 10 FEET OF TAPE. WRITE THE ABOVE THREE WORDS 4096(10) TIMES.

- B. WRITE THE BELOW THREE WORDS (SEE C) OF EXPAND CODE 99 TIMES.
C. EXPAND CODE. THREE WORDS OF EXPAND CODE SHOULD IMMEDIATELY FOLLOW EACH BLOCK.

0404
0404
0404

IN THE MARK TRACK THE WORDS APPEAR AS 010101010101 (2525(8)).

- D. THE FORWARD BLOCK MARK AND REVERSE GUARD REQUIRE THREE WORDS.

0404
4004
4040

WHICH APPEAR ON THE MARK TRACK AS 010110011010 (2632(8)).

- E. THE LOCK MARK, REVERSE CHECKSUM, REVERSE FINAL, REVERSE PREFINAL CONSIST OF SIX PDP-8 MEMORY WORDS.

0040
0000
4000
0040
0000
4000

THESE WORDS APPEAR ON THE MARK TRACK AS 001000001000001000001000 (10101010(8))

- F. MARK TRACK CODE FOR DATA IS GENERATED BY

4440
0044
4000

THESE THREE WORDS APPEAR AS 111000111000 (7070(8)) AND ARE REPEATED 41(10) TIMES FOR A 120 WORD BLOCK.

- G. THE PREFINAL, FINAL, CHECKSUM, AND REVERSE LOCK CONSIST OF SIX PDP-8 WORDS.

4440

4444
4044
4440
4444
4044

THESE WORDS APPEAR ON THE MARK TRACK AS 1110111110111101111011
(73737373(8)).

H. THE GUARD AND REVERSE BLOCK MARK CONSIST OF THREE WORDS

4040
0440
0404

WHICH APPEAR AS 101001100101 (5145(8)).

I. GENERATE 2702(8) BLOCK PATTERNS. REPEAT C THROUGH H. 2702(8)
TIMES.

J. 100 EXPAND CODES (SEE C.)

K. THE END ZONE PATTERN CONSIST OF THREE WORDS.

0400
4004
0040

WHICH APPEARS ON THE MARK TRACK AS 010010010010 (2222(8)).
REPEAT THESE 3 WORDS 4096(10) TIMES, SEE FIGURE 2 FOR A
DIAGRAM OF THE MARK TRACK AND DATA TRACKS.

Figure 1 Reading of the Mark Track

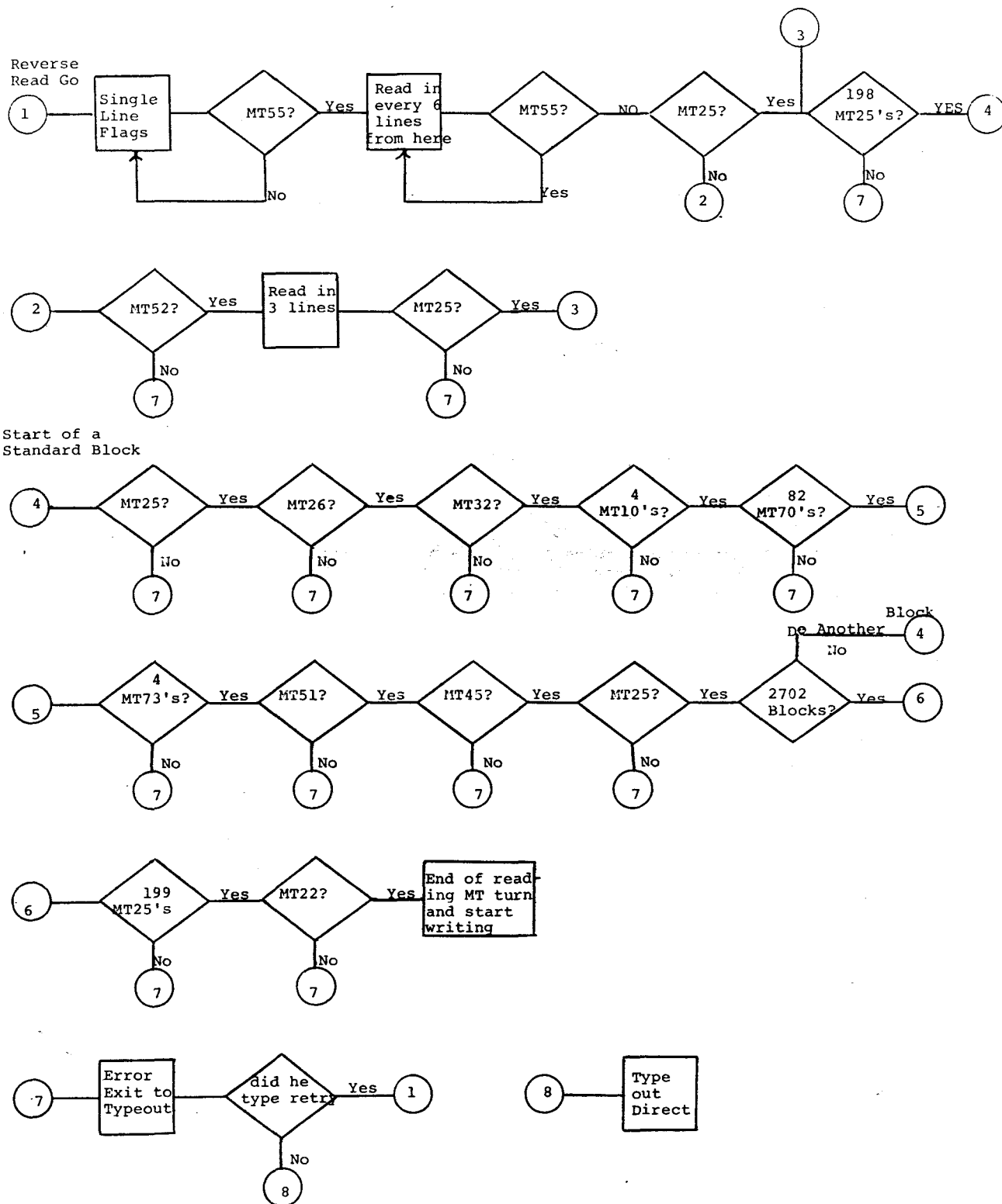


Figure 2 Mark Track and Data Track

Mark Track Name	End Zone					Expand Code					Forward Block Number					Reverse Guard					Lock Mark					Reverse Checksum					
Mark Track Code	1	0	1	1	0	1	0	1	0	1	0	1	0	1	0	0	0	1	0	0	0	0	0	1	0	0	0				
Data Track 0													0	0	0	0											0	0			
Data Track 1																														0	0
Data Track 2													0	0	0	0												0	0		
Line Flag	1	2	3	4	5	6	1	2	3	4	5	6	1	2	3	4	5	6	1	2	3	4	5	6	1	2	3	4	5	6	

[illegible][illegible]

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0010 X1=10
0011 X2=11

/SYMBOL TABLE AUGMENTATION

6771 SDSS=6771
6772 SDST=6772
6773 SDSQ=6773
6774 SDLC=6774
6775 SDLD=6775
6776 SDRC=6776
6777 SDRD=6777

0000	0000	*0	
0000	0000	0	
0001	5001	JMP	1 /HLT PROGRAM GOT INTERRUPTED SOMEHOW
0002	0002	2	
0003	0003	3	
0004	0000	0	
0005	0000	0	

/WORKING LOCATIONS

0020	0020	*20	
0020	0000	W1,	0000
0021	0000	W2,	0000
0022	0000	W3,	0000
0023	0000	W4,	0000
0024	0000	W5,	0000
0025	0000	W6,	0000
0026	0000	BLOCKS,	0000
0027	0000	DTA,	0000
0030	0000	PHASE,	0000
0031	0000	TOTAL,	0000
0032	0000	VAR1,	0000
0033	0000	VAR2,	0000

/CONSTANTS

0034	0017	C0017,	0017
0035	0070	C0070,	0070
0036	0077	C0077,	0077
0037	0007	C0007,	0007

0040	0700	C0700,	0700
0041	0203	C203,	0203
0042	0201	C201,	0201
0043	0260	C260,	0260
0044	0261	C261,	0261
0045	0270	C270,	0270
0046	0271	C271,	0271
0047	0277	C277,	0277
0050	1620	C1620,	1620
0051	7000	C7000,	7000
0052	7700	C7700,	7700
0053	7714	C7714,	7714
0054	7761	C7761,	7761
0055	0215	CRCOD,	0215
0056	0313	LETK,	0313
0057	0212	LFCOD,	0212
0060	7776	M2,	-2
0061	7775	M3,	-3
0062	7772	M6,	-6
0063	7771	M7,	-7
0064	7764	M14,	-14
0065	7634	M144,	-144
0066	0240	SPCOD,	0240

0067	3377	BADD,	BUFFER-1
0070	3400	BFR,	BUFFER
0071	0312	COMPAR,	COMPRE
0072	1055	IT,	INIT1
0073	0400	QU1,	Q1
0074	0410	QU2,	Q2
0075	0422	QU3,	Q3
0076	0434	QU4,	Q4
0077	0454	MESS,	MES
0100	1000	STX,	START
0101	0336	TYOCT,	TYCT
0102	0202	TYPE,	MESSAGE
0103	0260	TYPIN,	TYPN
0104	0600	WAIT,	STALL
0105	0000	WC,	0
0106	0000	MTR,	0
0107	3120	SLRDRG,	SRDRG
0110	0000	DATRD,	0
0111	7723	M55,	-55
0112	7753	M25,	-25
0113	7752	M26,	-26
0114	7746	M32,	-32
0115	7770	M10,	-10
0116	7710	M70,	-70
0117	7705	M73,	-73
0120	7727	M51,	-51
0121	7733	M45,	-45
0122	7756	M22,	-22
0123	7635	M143,	-143
0124	7726	M52,	-52

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0125 7747 M31, -31
0126 7472 M306, -306
0127 0020 CNT, 0
0130 7774 M4, -4
0131 7471 M307, -307
0132 3000 SSDSQT, SDSQT
0133 3027 SA3LNS, A3LNS
0134 3056 SCEXPC, CEXPC
0135 0077 MSK77, 0077
0136 3133 NUD, NUDTA
0137 0000 BLK, 0
0140 0000 REVBLK, 0
0141 3070 BCXOR, SBCXOR
0142 0000 CHKSUM, 0
0143 0000 SBWORD, 0

```

/TYPE THE CHARACTER IN THE AC ON THE KEYBOARD PRINTER

```

0144 0000 RSEND, 0000
0145 6046 TLS /LOAD AND PRINT, CLEAR FLAG
0146 6041 TSF /WAIT FOR CONFIRMATION
0147 5146 JMP I,-1 /ENDLESSLY
0150 6042 TCF /CLEAR THE FLAG ANYWAY
0151 5544 JMP I RSEND

```

/PRINT A "?" ON THE KEYBOARD TYPER

```

0152 0153 QU, I,+1
0153 6002 IOF
0154 7300 CLA CLL /C(AC)+C(L)=0
0155 1047 TAD C277 /"?"
0156 4144 JMS RSEND /TYPE THE CHARACTER
0157 5560 JMP I ,+1 /RESTART
0160 1061 INIT

```

/DECTAPE CONTROL WORDS

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0161 1400 DT1400, 1400
0162 0400 DT0400, 0400
0163 2000 DT2000, 2000
0164 3000 DT3000, 3000
0165 1000 DT1000, 1000

0166 3155 BINCO, BINCON
0167 0724 SELTIM, ZTIM
0170 0657 MARKER, ZMKT
0171 0613 BLKERR, ZBLK
0172 0637 DATERR, ZDATA
0173 0702 CHKERR, ZPAR
0174 1400 DOMARK, STMK

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0200      *200      /PAGE 1
          /TYPE CANNED MESSAGES,...,
          /THANKS TO DIGITAL 8-18-U
0220 5601      JMP I ,+1
0201 7515      PATCH

0222 0000      MESSAGE, 0
0203 6002      IOF
0204 7240      CLA      CMA      /SET C(AC)=-1
0205 1202      TAD      MESSAGE /ADD LOCATION
0206 3010      DCA      10      /AUTO INDEX REGISTER
0207 1410      TAD I 10      /FETCH FIRST WORD
0210 3221      DCA      MSRGHT /SAVE IT
0211 1221      TAD      MSRGHT
0212 7012      RTR
0213 7012      RTR      /ROTATE 6 BITS TO THE RIGHT
0214 7012      RTR
0215 4222      JMS      TYPECH /TYPE IT
0216 1221      TAD      MSRGHT /GET DATA AGAIN
0217 4222      JMS      TYPECH /TYPE RIGHT HALF
0220 5207      JMP      MESSAGE+5 /CONTINUE
0221 0000      MSRGHT, 0      /TEMPORARY STORAGE
0222 0000      TYPECH, 0      /TYPE CHARACTER IN C(AC)6-11
0223 0036      AND      C0077
0224 7450      SNA      /IS IT END OF MESSAGE?
0225 5410      JMP I 10      /YES: EXIT
0226 1253      TAD      M40      /SUBTRACT 40
0227 7500      SMA      /<40?
0230 5233      JMP      ,+3      /NO
0231 1254      TAD      C340      /YES: ADD 300
0232 5246      JMP      MTP      /TO CODES <40
0233 1061      TAD      M3      /SUBTRACT 3
0234 7440      SZA      /IS IT ZERO?
0235 5240      JMP      ,+3      /NO
0236 1255      TAD      C212      /YES: CODE 43 IS
0237 5246      JMP      MTP      /LINE-FEED (212)
0240 1060      TAD      M2      /SUBTRACT 2
0241 7440      SZA      /IS IT ZERO?
0242 5245      JMP      ,+3      /NO
0243 1256      TAD      C215      /YES: CODE 45 IS
0244 5246      JMP      MTP      /CARRIAGE RETURN (215)
0245 1257      TAD      C245      /ADD 200 TO OTHERS >40
0246 6046      MTP,      TLS      /TRANSMIT CHARACTER
0247 6041      TSF      /WAIT FOR THE FLAG
0250 5247      JMP      ,+1      /NOT SET YET
0251 7200      CLA      /SET! CLEAR C(AC)
0252 5622      JMP I TYPECH /RETURN

/CONSTANTS

0253 7740      M40,      -40
0254 0340      C340,     340
0255 0212      C212,     212

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0256 0215 C215, 215
0257 0245 C245, 245

/ROUTINE WAITS UNTILL A COMPLETE MESSAGE HAS BEEN ENTERED
/SIGNIFIED BY A CR.

0260 0000 TYPN, 0
0261 6002 IOF
0262 6032 KCC /CLEAR AC, KEYBOARD FLAG
0263 1067 TAD BADD /GET BUFFER ADDRESS
0264 3020 DCA W1 /STORE FOR THE CHARACTER STRING

/READ AND RESPOND WITH THE CHARACTER

0265 2020 NTYRTN, ISZ W1 /NORMAL RETURN, INCREMENT BUFFER
0266 6031 KSF /WAIT FOR KEYBOARD
0267 5266 JMP , -1 /FLAG TO RAISE
0270 6036 KRB /GOT FLAG, RESET IT, GET CHARACTER
0271 4144 JMS RSEND /SEND CHARACTER BACK
0272 3420 DCA I W1 /LOAD CHARACTER INTO BUFFER AREA

/IF CHARACTER IS A SPACE, IGNORE IT

0273 1420 TAD I W1 /CHARACTER INTO THE AC
0274 7041 CIA /SUBTRACT FROM SPACE CODE (240)
0275 1066 TAD SPCOD /COMPLETE COMPARISON
0276 7650 SNA CLA /HAS IT A SPACE?
0277 5266 JMP NTYRTN+1 /YES! DO NOT INCREMENT BUFFER

/IF CHARACTER IS A CR, EXIT FROM ROUTINE

0300 1420 TAD I W1 /CHARACTER TO AC
0301 7041 CIA /SET AC TO SUBTRACT CR (215)
0302 1055 TAD CRCOD /COMPLETE COMPARISON
0303 7640 SNA CLA /HAS IT CR?
0304 5265 JMP NTYRTN /NO! INCREMENT BUFFER & WAIT

/CARRIAGE RETURN FOUND, EXIT FROM ROUTINE

0305 1057 TAD LPCOD /GIVE KEYBOARD LINE FEED
0306 4144 JMS RSEND /EXECUTE LINE FEED
0307 7300 CLA CLL /EXIT WITH C(ACC) & AND C(L)=0
0310 6002 IOF
0311 5660 JMP I TYPN /RETURN TO CALL

/COMPARE A STRING OF CHARACTERS IN "BUFFER"
/TO A CHARACTER STRING AFTER A JMS IN ASCII

0312 0000 COMPRE, 0
0313 7240 CLA CMA /C(AC)=7777
0314 1312 TAD COMPRE /SUBTRACT 1 FOR INDEX REG I
0315 3010 DCA I0 /AUTO INDEX 1 SET TO CMA STRING
0316 1067 TAD BADD /AUTO INDEX 2 SET TO BUFFER-1
0317 3011 DCA I1 /LOAD X2

/COMPARE CHARACTERS TILL ONE DOESN'T COMPARE OR TILL
/A 0 IS FOUND IN X1. IF OK, RETURN TO TWO PLUS THE
/ZERO, IF BAD ONE PLUS

0320	1410	TAD I X1	/CHARACTER FROM PROGRAM
0321	7041	CIA	/TO SUBTRACT FROM
0322	1411	TAD I X2	/CHARACTER IN BUFFER
0323	7640	SZA CLA	/COMPARE?
0324	5332	JMP CERR	/NOIRESYNC FOR NON COMPARE EXIT
0325	1410	TAD I X1	/YES! CHECK FOR GOOD EXIT
0326	7440	SZA	/IF 0, EXIT GOOD
0327	5321	JMP -6	/NO! TEST NEXT CHARACTER
0330	2010	ISZ X1	/+1 TO X1(TOTAL 2 FROM THE 0)
0331	5410	JMP I X1	/+1 TO X1, EXIT

/ERROR FOUND. RESYNC AND EXIT NO COMPARE

0332	1410	CERR, TAD I X1	/CHARACTER FROM PROGRAM
0333	7640	SZA CLA	/IS THIS EXIT KEY? (0000)
0334	5332	JMP -2	/NO! GET NEXT
0335	5410	JMP I X1	/YES! EXIT, NOT COMPARE

/TYPE ONE FOUR CHARACTER OCTAL WORD GIVEN TO THE
/ROUTINE VIA C(ACC), C(ACC)=0 ON EXIT

0336	0000	TYCT, 0	
0337	3376	DCA TH1	/STORE WORD GIVEN
0340	1376	TAD TH1	/TO C(ACC) AGAIN
0341	7012	RTR	
0342	7012	RTR	/6 BITS RIGHT
0343	7012	RTR	
0344	3373	DCA TYCTI+2	/SAVE ROTATED VALUE, 1ST TWO
0345	1373	TAD TYCTI+2	/TO C(ACC) AGAIN
0346	0037	AND C0007	/ISOLATE SECOND CHARACTER
0347	1377	TAD C0060	/CONVERT TO ASCII
0350	3372	DCA TYCTI+1	/STORE AS FIRST PARTIAL 2
0351	1373	TAD TYCTI+2	/ROTATED VALUE STORED ABOVE
0352	7006	RTL	
0353	7004	RAL	/3 BITS LEFT
0354	0040	AND C0700	/ISOLATE FIRST CHARACTER
0355	1372	TAD TYCTI+1	/CONVERT 1ST TO ASCII
0356	3372	DCA TYCTI+1	/1ST AND 2ND CHARACTERS READY
0357	1376	TAD TH1	/ORIGIONAL WORD
0360	0037	AND C0007	/ISOLATE 4TH CHARACTER
0361	1377	TAD C0060	/CONVERT 4 TH TO ASCII
0362	3373	DCA TYCTI+2	/STORE 4TH FOR A MOMENT
0363	1376	TAD TH1	/ORIGIONAL WORD
0364	7006	RTL	
0365	7004	RAL	/POSITION IT 3RD CHARACTER
0366	0040	AND C0700	/ISOLATE 3RD CHARACTER
0367	1373	TAD TYCTI+2	/CONVERT TO ASCII
0370	3373	DCA TYCTI+2	/CONVERSION COMPLETE
0371	4502	TYCTI, JMS I TYPE	/TYPE THE FOUR CHARACTERS
0372	0000	0	/FIRST 2
0373	0000	0	/SECOND 2

0374 0000 0 /KILL KEY
0375 5736 JMP I TYCT /EXIT FROM ROUTINE

/SOME CONSTANTS FOR THE ROUTINE

0376 0000 TW1, 0000
0377 6060 C6060, 6060

0400 *400
/VARIOUS ERROR MESSAGES
/"NOT DECIMAL"

0400 4502 Q1, JMS I TYPE
0401 1617 1617 /NO
0402 2440 2440 /T
0403 0405 0405 /OE
0404 0311 0311 /CI
0405 1501 1501 /MA
0406 1400 1400 /L
0407 5247 JMP GUX

/"TO MANY WORDS"

0410 4502 Q2, JMS I TYPE
0411 2417 2417 /TO
0412 1740 1740 /O
0413 1501 1501 /MA
0414 1631 1631 /NY
0415 4027 4027 /W
0416 1722 1722 /OR
0417 0423 0423 /DS
0420 0000 0000 /00
0421 5247 JMP GUX

/"TO MANY BLOCKS"

0422 4502 Q3, JMS I TYPE
0423 2417 2417 /TO
0424 1740 1740 /O
0425 1501 1501 /MA
0426 1631 1631 /NY
0427 4002 4002 /B
0430 1417 1417 /LQ
0431 0313 0313 /CK
0432 2300 2300 /S0
0433 5247 JMP GUX

/"NOT DIVISIBLE BY 3"

0434 4502 Q4, JMS I TYPE
0435 1617 1617 /NO
0436 2440 2440 /T
0437 0411 0411 /DI
0440 2611 2611 /VI
0441 2311 2311 /SI
0442 0214 0214 /BL

```

0443 0540      0540 /E
0444 2231      0231 /BY
0445 4063      4063 / 3
0446 0000      0000 /00
0447 4502      GUX, JMS I TYPE
0450 4345      4345 /CR+LF
0451 0000      0000 /END
0452 5653      JMP I ,+1
0453 1061      INIT
    
```

/THE CODING BELOW CREATES THE BLOCK NUMBER
/CONVERSION PRIOR TO THE TAPE WRITE.

```

0454 0000      MES, 0
0455 3023      DCA W4 /SAVE WORD
0456 7100      CLL
0457 1023      TAD W4
0460 7052      CMA RTR
0461 7012      RTR
0462 0051      AND C7000
0463 3306      DCA V1
0464 1023      TAD W4
0465 7046      CMA RTL
0466 7004      RAL
0467 0040      AND C0700
0470 3307      DCA V2
0471 1023      TAD W4
0472 7052      CMA RTR
0473 7010      RAR
0474 0035      AND C0070
0475 3313      DCA V3
0476 1023      TAD W4
0477 7046      CMA RTL
0500 7006      RTL
0501 0037      AND C0007
0502 1306      TAD V1
0503 1307      TAD V2
0504 1313      TAD V3
0505 5654      JMP I MES

0506 0000      V1, 0000
0507 0000      V2, 0000
0510 7777      7777
0511 7700      7700
0512 0000      0000
0513 0000      V3, 0000
0514 0000      0000

0515 7200      PATCH, CLA
0516 1322      TAD ,+4
0517 3001      DCA 1
0520 5721      JMP I ,+1
0521 1000      START
0522 7402      HLT
    
```

0600 *600

```

0600 0000 STALL, 0
0601 7200 CLA
0602 1412 TAD I 12 /WORD TO BE WRITTEN
0603 6773 SOSQ /WAIT FOR QUADLINE FLAG
0604 5203 JMP ,+1
0605 6775 SOLC /LOAD DATA REGISTERS
0606 6772 SDST /CHECK FOR TIMING ERROR
0607 7410 SKP
0610 4567 JMS I SELTIM /TIMING ERROR
0611 7200 CLA
0612 5600 JMP I STALL /GO GET NEXT WORD

```

/WAIT TILL WORD COUNT REGISTER GOES TO ZERO
/BLOCK NUMBER ERROR

```

0613 0000 ZBLK, 0
0614 7200 CLA
0615 1027 TAD DTA
0616 6774 SOLC /STOP MOVEMENT OF TAPE
0617 4502 JMS I TYPE
0620 2003 /PC
0621 4000 /END
0622 7240 CLA CHA
0623 1213 TAD ZBLK
0624 4501 JMS I TYOCT
0625 4502 JMS I TYPE
0626 4040 /DOUBLE SPACE
0627 0214 /BL
0630 1703 /OC
0631 1340 /K
0632 1625 /NU
0633 1502 /MB
0634 0522 /ER
0635 4000 /END
0636 5344 JMP ZCOM

```

/DATA ERRORS

```

0637 0000 ZDATA, 0
0640 7200 CLA
0641 1027 TAD DTA
0642 6774 SOLC /STOP THE TAPE
0643 4502 JMS I TYPE
0644 2003
0645 4000
0646 7240 CLA CHA
0647 1237 TAD ZDATA
0650 4501 JMS I TYOCT
0651 4502 JMS I TYPE
0652 4040
0653 0401 /DA
0654 2401 /TA
0655 4000 /END

```

```

0656 5344      JMP      ZCOM

/ MARK TRACK ERROR

0657 0000      ZMKTk, 0
0660 7200      CLA
0661 1027      TAD      DTA
0662 6774      SOLC      /STOP THE TAPE
0663 4502      JMS I TYPE
0664 2003      2003      /PC
0665 4000      4000      /END
0666 7240      CLA      CMA
0667 1257      TAD      ZMKTk
0670 4501      JMS I TYOCT
0671 4502      JMS I TYPE
0672 4040      4040
0673 1501      1501      /MA
0674 2213      2213      /RK
0675 4024      4024      / T
0676 2201      2201      /RA
0677 0313      0313      /CK
0700 4000      4000      / 0
0701 5344      JMP      ZCOM

```

/PARITY ERROR

```

0702 0000      ZPAR, 0
0703 7200      CLA
0704 1027      TAD      DTA
0705 6774      SOLC      /STOP THE TAPE
0706 4502      JMS I TYPE
0707 2003      2003      /PC
0710 4000      4000      /END
0711 7240      CLA      CMA
0712 1302      TAD      ZPAR
0713 4501      JMS I TYOCT
0714 4502      JMS I TYPE
0715 4040      4040
0716 0310      0310      /CH
0717 0503      0503      /EC
0720 1323      1323      /KS
0721 2515      2515      /UM
0722 4000      4000      /0
0723 5344      JMP      ZCOM

```

/TIMING ERROR

```

0724 0000      ZTIM, 0
0725 7200      CLA
0726 1027      TAD      DTA
0727 6774      SOLC      /STOP THE TAPE
0730 4502      JMS I TYPE
0731 2003      2003
0732 4000      4000

```

```

0733 7240      CLA      CMA
0734 1324      TAD      ZTIM
0735 4501      JMS I TYOCT
0736 4502      JMS I TYPE
0737 4040      4040
0740 2411      2411 /TI
0741 1511      1511 /MI
0742 1607      1607 /NG
0743 4000      4000 / 0

```

/TYPE "ERROR PHASE X"

```

0744 1030      ZCOM,   TAD   PHASE   /WHAT PHASE OF OPERATION
0745 1363      TAD   PFORM   /WAS THE MACHINE IN
0746 3356      DCA   TFORM   /WHEN ERROR OCCURED
0747 4502      JMS I TYPE
0750 0522      0522   /ER
0751 2217      2217   /RO
0752 2240      2240   /R
0753 2010      2010   /PH
0754 0123      0123   /AS
0755 0540      0540   /E
0756 4060      TFORM,  4060   / X
0757 4543      4543   /CR+LF
0760 0000      0000   /END
0761 5762      JMP I   .+1
0762 2726      RETRY
0763 4060      PFORM,  4060

```

/HERE STARTS THIS PROGRAM, IT WILL ASK THE
/OPERATOR FOR DRIVE NUMBERS, THEN ASK HIM FOR
/A DIRECTION ON WHAT TO DO WITH THE DRIVES.

/THE SEQUENCE FOR MARKING A TAPE WOULD APPEAR AS:

/UNIT? (0 OR 1 OR 0 1)
/FORMAT? (MARK 1215)
/2277 WORDS, 0256 BLOCKS, OK? YES OR NO
/(YES)

/THAT DATA IN PARENTHESIS IS TYPED BY THE OPERATOR
/(HE DOESN'T TYPE THE PARENTHESIS)
/IF HE HAD ANSWERED NO, "FORMAT?" WOULD BE TYPED OUT.
/IF THE DRIVE WAS WRONG, HE WOULD TYPE RESTART.
/IF HE HAD TYPED "MARK" IN RESPONSE TO "FORMAT?" THE
/TAPE WOULD BE MARKED WITH THE STANDARD PDP-8 CONFIGURATION.
/IF HE HAD TYPED "MARK 384" THE TAPE WOULD
/BE MARKED WITH THE STANDARD PDP-10 CONFIGURATION
/NOTE! THE WORD AND BLOCK NUMBERS ARE TYPED IN OCTAL
/IF A MISTAKE OCCURS ON THE OPERATORS PART (WITH REFERENCE
/TO BLOCK + WORD SIZE) HE WILL BE TOLD ABOUT IT

```

1020      *1020

/MAKE A CALL FOR THE DECTAPE NUMBERS TO BE
/WORKED.

1000 4502  START,  JMS I TYPE      /SET UP TYPER
1021 4543          4543 /CR+LF
1022 4300          4300 /LF+END

1003 4502  TYQU,   JMS I TYPE      /"UNIT?"
1024 2516          2516 /UN
1005 1124          1124 /IT
1006 7740          7740 /?
1027 0000          0000 /END

/WAIT FOR A REPLY

1010 4503          JMS I TYPIN     /GET NUMBERS
1011 1067          TAD      BADD     /INITIALIZE POINTER (BFR)
1012 7001          IAC      BADD     / (BADD=BUFFER-1, SO BUMP THE AC)
1013 3070          DCA      BFR      /TO START OF INPUT BUFFER
1014 3374          DCA      DCTR     /INITIALIZE DTA COUNTER TO 0
1015 3346          DCA      CRFLAG   /CLEAR FLAG SO CR NOT ACCEPTIBLE
1016 1055          CRCHK, TAD      CRCOD /GET CODE FOR CAR, RETN
1017 7041          CIA          /NEGATE IT
1020 1470          TAD I   BFR      /SEE IF NEXT CHAR. IN
1021 7450          SNA          /BUFFER IS CAR, RETN.
1022 5244          JMP      OKCR     /YES! SEE IF C,R, LEGAL HERE
1023 3346          DCA      CRFLAG   /NO! SO C,R, IS LEGAL NOW
1024 1043          VALCHK, TAD      C260 /SEE IF # IS LESS THAN
1025 7041          CIA          /ASCII 0 (260)
1026 1470          TAD I   BFR      /SUBTRACT BUFFER DATA
1027 7710          SPA      CLA      /IS IT LESS THAN ASCII 0?
1030 5203          JMP      TYQU     /YES! TELL OUTSIDE WORLD
1031 1044          TAD      C261     /NO! SEE IF GREATER THAN
1032 7040          CMA          /ASCII 1 (261)
1033 1470          TAD I   BFR      /SUBTRACT BUFFER DATA
1034 7700          SMA      CLA      /GREATER THAN ASCII ??
1035 5203          JMP      TYQU     /YES! TELL OUTSIDE WORLD
1036 1470          TAD I   BFR      /NO! ACCEPT BUFFER
1037 7012          RTR
1040 0051          AND      C7000    /ISOLATE DTA
1041 4347          JMS      REPEAT   /GO CHECK FOR REPEATED DTA AND STORE #
1042 2070          ISZ      BFR      /INCREMENT INPUT BUF. PTR.
1043 5216          JMP      CRCHK    /GO LOOK AT NEXT CHAR.

/THIS SECTION CHECKS TO SEE IF THERE HAS BEEN ANY
/VALID INPUT ONCE A CARRIAGE RETURN IS SEEN
1044 7200          OKCR,  CLA      /CLEAR AC

```

1045	1346	TAD	CRFLAG	/LOAD CR FLAG; 0 MEANS NO GOOD
1046	7650	SNA	CLA	
1047	5220	JMP	START	/01 NO VALID INPUT; RESTART
1050	1374	TAD	DCTR	/NOT 01 SO HAVE VALID INPUT
1051	1376	TAD	DBUFAD	/CALCULATE END OF DTA LIST +1
1052	3375	DCA	DBUFPT	/STORE IT IN BUFFER POINTER, THEN
1053	7040	CMA		/COMPLEMENT THE AC AND
1054	3775	DCA I	DBUFPT	/TERMINATE DTA LIST WITH 7777
1055	7200	INIT1,	CLA	/CLEAR AC IF COME THRU LOC IT
1056	1376	TAD	DBUFAD	/AND RESET LIST POINTER
1057	3375	DCA	DBUFPT	/TO START OF LIST
1060	4745	JMS I	GETDTA	/GO GET A DTA NUMBER

/INFORM THE OPERATOR THAT THE PROGRAM IS SET TO START
/TYPE "FORMAT" AND WAIT FOR THE REPLY

1061	4502	INIT,	JMS I TYPE	/MESSAGE OUT
1062	0617		0617	/FO
1063	2215		2215	/RM
1064	0124		0124	/AT
1065	7740		7740	/?
1066	0000		0000	/END
1067	4503		JMS I TYPIN	/WAIT FOR A REPLY
1070	4471		JMS I COMPAR	/DID HE TYPE "MARK"?
1071	0315		0315	/M
1072	0301		0301	/A
1073	0322		0322	/R
1074	0313		0313	/K
1075	0000		0000	/END
1076	5301		JMP	,+3
1077	5700		JMP I	,+1
1100	1200		MARK	/TO MARK A TAPE

/SEE IF HE TYPED "RDR" (READ AND TYPE FIRST 12
/BLOCK NUMBERS IN REVERSE).

1101	4471		JMS I COMPAR	
1102	0322		0322	/R
1103	0304		0304	/D
1104	0322		0322	/R
1105	0000		0000	/0
1106	5311		JMP	,+3
1107	5710		JMP I	,+1
1110	2677		RDR	/TYPE BLOCKS

/SEE IF HE TYPED "RDF" (READ AND TYPE FIRST 12
/BLOCK NUMBERS FORWARD).

1111	4471		JMS I COMPAR	
1112	0322		0322	/R
1113	0304		0304	/D
1114	0306		0306	/F
1115	0000		0000	/0
1116	5321		JMP	,+3
1117	5720		JMP I	,+1

1120 2600 RDFA /TYPE BLOCKS

/SEE IF HE TYPED "SAME" (MEANING MARK A TAPE
/USING THE SAME CONSTANTS AS BEFORE),

1121 4471 JMS I COMPAR
1122 7323 0323 /S
1123 0301 0301 /A
1124 0315 0315 /M
1125 2305 0305 /E
1126 0000 0000 /0
1127 5332 JMP .+3
1130 5731 JMP I .+1
1131 3200 SWCHK /TO MARK AS BEFORE

/SEE IF HE TYPED "RESTART"

1132 4471 JMS I COMPAR
1133 0322 0322 /R
1134 0305 0305 /E
1135 0323 0323 /S
1136 0324 0324 /T
1137 0301 0301 /A
1140 0322 0322 /R
1141 0324 0324 /T
1142 0000 0000 /0
1143 4152 JMS GU /MUST BE NONSENSE
1144 5200 JMP START /START ALL OVER
1145 3133 GETDTA, NUDTA /POINTER TO ROUTINE TO SWITCH UNITS
1146 0000 CRFLAG, 0 /#0, CR NO GOOD; NOT 0, CR IS OK

/SUBROUTINE TO CHECK FOR REPEATED DTA NUMBERS
/DTA # TO COMPARE TO LIST IS IN AC ON ENTRY--THIS
/ROUTINE STORES THE DTA # IF IT IS NEW AND IGNORES IT
/IF IT IS NOT--CALL BY JMS REPEAT WITH DTA # IN AC
REPEAT, 0

1147 0000 DCA DNUM /TEM STORAGE FOR NEW DTA #
1150 3377 TAD DBUFAD /INITIALIZE POINTER (DBUFPT)
1151 1376 DCA DBUFPT /TO START OF DTA LIST
1152 3375 TAD DCTR /LOAD NUM. OF DTAS STORED
1153 1374 CMA /COMPLEMENT IT
1154 7040 DCA COMCTR /STORE IN COMPARE COUNTER
1155 3373 COMCHK, ISZ COMCTR /DONE WITH ALL COMPARES?
1156 2373 JMP DCOMP /NO; GO DO COMPARE
1157 5364 TAD DNUM /YES; STORE NEW DTA#
1160 1377 DCA I DBUFPT /AT END OF LIST
1161 3775 ISZ DCTR /INCR. # OF DTAS STORED
1162 2374 JMP I REPEAT /RETURN
1163 5747

/THIS SECTION DOES THE ACTUAL COMPARISON BETWEEN
/THE DTA# PASSED TO THE ROUTINE AND A NUMBER ON THE LIST

1164 1775 DCOMP, TAD I DBUFPT /GET NEXT DTA NUMBER FROM LIST
1165 7041 CIA /NEGATE IT
1166 1377 TAD DNUM /ADD IN DTA NUMBER PASSED

```

1167 7650      SNA   CLA   /ARE THEY THE SAME?
1170 5747      JMP I  REPEAT /YES! RETURN
1171 2375      ISZ   DBUFPT /NO! INCREMENT LIST POINTER
1172 5356      JMP   COMCHK /SEE IF DONE ALL COMPARES
/
/
1173 0000      COMCTR, 0      /COUNTER FOR # OF LIST COMPARISONS TO BE DONE
1174 0000      DCTR, 0       /COUNTER FOR # OF DTAS IN LIST
1175 0000      DBUFPT, 0     /POINTER TO CURRENT POSITION IN DTA LIST
1176 3162      DBUFAD, DTABUF /START OF DTA NUM, LIST
1177 0000      DNUM, 0       /TEM STORAGE FOR DTA #
/
/
PAUSE

1200          *1200
/ MARK WAS TYPED IN, IF W1=1 IS NOT A "K", ASSUME THAT
/ A NUMBER WAS TYPED IN, AND VERIFY THIS, IF W1=1 IS
/ A "K", ASSUME STANDARD FORMAT, (W1=LAST ENTRY INTO THE BUFFER)

1200 1566      MARK, TAD I BINCO /ADDRESS OF FIRST BINARY
1201 3024      DCA W5          /CONSTANT FOR DEC TO BIN
1202 3031      DCA TOTAL      /WILL BE BINARY EQUIVILANT

/SAVE C(X1) FOR DECREMENT THROUGH BUFFER

1203 7240      DNC, CLA CMA    /DECREMENT BUFFER ADDRESS
1204 1020      TAD W1         /ADDRESS BY 1
1205 3020      DCA W1         /W1=SWEPT ADDRESS

/LOOK FOR END OF PROCESSING BY LOOKING FOR A "K" IN BUFFER

1206 1056      TAD LETK      /LETTER ASCII "K"
1207 7041      CIA          /SUBTRACT FROM CHARACTER
1210 1420      TAD I W1      /IN BUFFER
1211 7650      SNA CLA      /EQUAL?
1212 5244      JMP DIV3      /YES! SEE IF DIVISIBLE BY 3

/VERIFY THIS CHARACTER AS BEING OF DECIMAL ORIGIN

1213 1043      TAD C260      /ASCII FOR 0
1214 7041      CIA          /TO SEE IF CHARACTER
1215 1420      TAD I W1      /IS LESS THAN 260
1216 7710      SPA CLA      /IS IT?
1217 5473      JMP I OU1     /YES! NOT DECIMAL CHARACTER
1220 1046      TAD C271      /ASCII FOR 9
1221 7040      CMA          /TO SEE IF GREATER THAN
1222 1420      TAD I W1      /9
1223 7700      SNA CLA      /IS IT?
1224 5473      JMP I OU1     /NOT A DECIMAL CHARACTER

/CHARACTER IS DECIMAL, NOW CONVERT IT TO BINARY
/REMEMBER POSITION OF CHARACTER IN BUFFER MAY BE
/10,100,1000.

```

```

1225 1420      TAD I W1      /ISOLATE THE NUMBER
1226 0034      AND C0017     /FOR PROPER CONVERSION
1227 7450      SNA          /IF 0, NO BINARY CONVERSION NEEDED
1230 5242      JMP IBS       /YES! 0: INCREMENT BINARY CONVERSION

```

- /NOT 0, SET UP CONVERSION LOOP

```

1231 7141      CLL CIA       /NUMBER OF ADDITIONS
1232 3023      DCA W4        /TO NEGATIVE FOR ISZ
1233 1424      TAD I W5      /BINARY POSITION TO C(ACC)
1234 1031      TAD TOTAL     /ADD TO PRESENT TOTAL
1235 7430      SZL          /CHECK ON TO MANY WORDS
1236 5474      JMP I GU2     /TO MANY WORDS CALLED FOR
1237 3031      DCA TOTAL     /KEEP RUNNING SUM
1240 2023      ISZ W4        /LAST ADDITION?
1241 5233      JMP , -6      /NO: ADD AGAIN

```

/FINAL ADDITION FOR THIS POSITION COMPLETED

```

1242 2024      IBS, ISZ W5    /NEXT POSITION
1243 5223      JMP DMC       /DO NEXT CHARACTER

```

/LAST CHARACTER COMPLETED, SEE IF DIVISIBLE BY 3
/IF NOT A NORMAL INPUT

```

1244 1031      DIV3, TAD TOTAL /GET TOTAL WORDS
1245 7450      SNA          /IF TOTAL 0, NORMAL INPUT
1246 1042      TAD C201      /129 OCT. THIS TEST REDUNDANT
1247 1034      TAD C0017     /ADD CONSTANT 15 TO TOTAL
1250 3031      DCA TOTAL     /FOR FUTURE CONSIDERATIONS
1251 3032      DCA VAR1      /# OF WORDS/3 FOR MARK TRACK WRITING
1252 1031      TAD TOTAL     /RESTORE IN THE ACC
1253 7100      CLL          /TO DIVIDE BY 3, LINK KEEPS OVERFLOW
1254 1061      TAD M3        /SUBTRACT 3
1255 2032      ISZ VAR1      /ON EACH DIVISION, KEEP RUNNING SUM
1256 7440      SZA          /IF AC = 0, NO REMAINDER
1257 7420      SNL          /WHEN LINK GOES TO 0, DIVISION ENDED
1260 7410      SKP          /NOW SEE IF IT DIVIDED EVENLY
1261 5253      JMP , -6      /SUBTRACT 3 MORE
1262 7640      SZA CLA      /IF 0, OK, OTHERWISE ERROR
1263 5476      JMP I GU4     /NOT DIVISIBLE BY 3

```

/CORRECT "VAR1" (THE NUMBER OF WORDS/3) FOR THE +15
/ADDED JUST ABOVE AND AN INHERANT +2 DUE TO MARK TRACK
/CONFIGURATION TO BE WRITTEN,

```

1264 1063      TAD M7        /SUBTRACT 7 FROM PRONY SETUP
1265 1032      TAD VAR1      /GIVING THE NUMBER OF TIMES
1266 7041      CIA          /TO BE USED LATER IN A ISZ
1267 3032      DCA VAR1      /DATA MARK WILL BE WRITTEN

```

/COMPUTE A VALUE FOR TOTAL NUMBER OF BLOCKS
/RECORD SIZE * 15 INTO 636160 OCT,

```

1270 1053      TAD C7714     /EXTENDED 64 VALUE, SETS AC#2

```

1271	3020	DCA	W1	/SET FOR 640000
1272	4755	JMS I	FORM10	/PATCH TO CHECK FOR STD.10 FORMAT
1273	1050	TAD	C1620	/VERNIER ADJUSTMENT FOR FORMULA
1274	7100	CLL		/ACC#2 CARRY FUNCTION
1275	1031	TAD	TOTAL	/WORD COUNT
1276	2026	ISZ	BLOCKS	/+1 TO BLOCK COUNT
1277	7410	SKP		
1300	5475	JMP I	OU3	/TO MANY BLOCKS CALLED FOR
1301	7420	SNL		/CARRY INTO ACC#2?
1302	5275	JMP	,=5	/NO: CONTINUE COUNT
1303	2020	ISZ	W1	/YES: FULLY DIVIDED?
1304	5274	JMP	,=-10	/NO: CONTINUE PROCESS
1305	7300	CLA	CLL	/C(ACC)+ C(L)=0
1306	1026	F10RTN, TAD	BLOCKS	/FOR MARK TRACK (COME HERE FR F10PAT IF 10 FRMT)
1307	7040	CMA		/WRITING
1310	3033	DCA	VAR2	/SEE MARK WRITE

/VALUES FOR BLOCK AND RECORD SIZE HAVE BEEN
/COMPUTED, TELL OUTSIDE WORLD AND GET THE OK.

1311	1031	TAD	TOTAL	/SUBTRACT 15 FROM TOTAL
1312	1054	TAD	C7761	/WORDS FOOLING OPERATOR
1313	3031	DCA	TOTAL	/CORRECTED FOR TAPE WRITING
1314	1031	TAD	TOTAL	/FOR OCTAL TYPEOUT
1315	4501	JMS I	TYOCT	/TYPE OCTAL WORDS
1316	4502	JMS I	TYPE	/TYPE MESSAGE
1317	4027		/ W	
1320	1722		/OR	
1321	0423		/DS	
1322	5400		//, END	
1323	1026	TAD	BLOCKS	/TYPE OUT BLOCK #S
1324	7001	IAC		/TO FOOL THE OPERATOR
1325	4501	JMS I	TYOCT	/IN OCTAL
1326	4502	JMS I	TYPE	/TYPE MESSAGES
1327	4002		/ B	
1330	1417		/LO	
1331	0313		/CK	
1332	2356		/S.	
1333	1713		/OK	
1334	7733		???	
1335	3105		/YE	
1336	2340		/S	
1337	1722		/OR	
1340	4016		/ N	
1341	1735		/O)	
1342	4543		/CR+LF	
1343	0000		/END	
1344	4503	JMS I	TYPIN	/WAIT FOR REPLY

/SEE IF A YES OR NO ANSWER WAS GIVEN

1345	4471	JMS I	COMPAR
1346	0331		/Y
1347	0305		/E
1350	0323		/S

```

1351 2000      0000 /END
1352 5472      JMP I IT

1353 5754      JMP I ,+1
1354 3230      SWCHK
1355 1556      FORM10, F10PAT

1400      *1437
          /SET THE TAPE INTO MOTION, ALL VARIABLES ARE SET.
          /WRITE TIMING AND MARK TRACK

1400 7200      STMK, CLA
1401 3030      DCA PHASE
1402 1161      TAD DT1400 /FWD, WRITE, GO
1403 1027      TAD DTA /GET UNIT NUMBER
1404 6774      SDLC /LOAD COMMAND REGISTER
1405 1033      TAD VAR2 /TO MAKE A RESTART FOR THE SAME
1406 3025      DCA W6 /OPTION POSSIBLE

          /WRITE ABOUT 10 FEET OF END ZONE

1407 3020      DCA W1
1410 1310      CEZ, TAD REZ /ADDRESS OF DATA
1411 4270      JMS SETUP
1412 2020      ISZ W1
1413 5210      JMP CEZ /NOT END FOOTAGE
1414 1065      TAD M144 /OK WRITE INTERBLOCK SYNC
1415 3020      DCA W1
1416 4222      JMS INBLSY
1417 2020      ISZ W1
1420 5216      JMP ,+2
1421 5230      JMP W0Z

          /WRITE INTERBLOCK SYNC

1422 0000      INBLSY, 0
1423 1032      TAD VAR1 /RESET THE WORDS
1424 3024      DCA W5
1425 1314      TAD IBZ /ADDRESS OF DATA
1426 4270      JMS SETUP /GO OUT AND WRITE 1
1427 5622      JMP I INBLSY /GO DO AGAIN

          /WRITE FORWARD BLOCKMARK AND REVERSE GUARD

1430 1320      W0Z, TAD PBM /ADDRESS OF PATTERN
1431 4270      JMS SETUP

          /WRITE LOCKMARK, REVERSE CHECKSUM, REV FINAL, REV PREFINAL

1432 1324      LRCFP, TAD WLMRF
1433 4300      JMS SETUP1

          /WRITE THE DATA TRACK

1434 1333      DTRK, TAD DZ /ADDRESS OF PATTERN
1435 4270      JMS SETUP
1436 2024      ISZ W5
1437 5234      JMP DTRK /NOW WRITE DATA MARK TRACK AGAIN

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1440 1337 PFCRC, /WRITE PREFINAL, FINAL, CHECKSUM, AND REVERSE LOCK
1441 4300 TAD FEZ /ADDRESS OF DATA
JMS SETUP1

1442 1346 GRB, /WRITE GUARD REVERSE BLOCK
1443 4270 TAD GRZ
JMS SETUP

/THIS COMPLETES 1 BLOCK, GO BACK AND WRITE THE REST
1444 4222 JMS INBLSY /WRITE INTERBLOCK SYNC
1445 2025 ISZ W6 /TOTAL NUMBER OF BLOCKS
1446 5230 JMP W0Z /WRITTEN? NO!

/ALL DATA BLOCKS WRITTEN NOW WRITE BUFFER ZONE OF INTERBLOCK SYNC
1447 1123 TAD M143 /198 EXPAND CODES AT END OF BLOCKS
1450 3020 DCA W1
1451 4222 JMS INBLSY
1452 2020 ISZ W1
1453 5251 JMP ,+2

/FINISHED BLOCK WRITTING, WRITE ANOTHER (0?1) OF END ZONES
1454 3020 DCA W1
1455 1352 WEZF, TAD EZM
1456 4270 JMS SETUP
1457 2020 ISZ W1
1460 5255 JMP WEZF
1461 6772 SDST
1462 7610 SKP CLA
1463 4567 JMS I SELTIM /TIMING ERROR
1464 1373 TAD C1
1465 3030 DCA PHASE
1466 5667 JMP I ,+1
1467 1600 MWTM

1470 0000 SETUP, 0
1471 3012 DCA I2 /WORD TO BE WRITTEN ON MARK TRACK
1472 1061 TAD M3
1473 3105 DCA WC
1474 4504 JMS I WAIT
1475 2105 ISZ WC
1476 5274 JMP ,+2
1477 5670 JMP I SETUP

1500 0000 SETUP1, 0
1501 3012 DCA I2
1502 1062 TAD M6
1503 3105 DCA WC
1504 4504 JMS I WAIT
1505 2105 ISZ WC
1506 5304 JMP ,+2
1507 5700 JMP I SETUP1

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/THESE ARE THE DATA CONFIGURATIONS FOR THE MARK TRACK

/REVERSE END ZONE

1510	1510	REZ,	.	
1511	4044		4044	/ON TAPE AS 5555 (OCT)
1512	0440		0440	
1513	4404		4404	

/INTERBLOCK SYNC

1514	1514	IBZ,	.	
1515	0404		0404	/ON TAPE AS 2525 (OCT)
1516	0404		0404	
1517	0404		0404	

/FORWARD BLOCK MARK AND REVERSE GUARD

1520	1520	FBM,	.	
1521	0404		0404	/ON TAPE AS 2632 (OCT)
1522	4004		4004	
1523	4040		4040	

/LOCK MARK, REVERSE CHECKSUM, REVERSE FINAL
/AND REVERSE PREFINAL

1524	1524	WLHRF,	.	
1525	0040		0040	/ON TAPE AS 10101010 (OCT)
1526	0000		0000	
1527	4000		4000	
1530	0040		0040	
1531	0000		0000	
1532	4000		4000	

/DATA MARK

1533	1533	DZ,	.	
1534	4440		4440	/ON TAPE AS 7070 (OCT)
1535	0044		0044	
1536	4000		4000	

/PREFINAL, FINAL, FWD CHECKSUM, AND REVERSE LOCK

1537	1537	FEZ,	.	
1540	4440		4440	/ON TAPE AS 73737373 (OCT)
1541	4444		4444	
1542	4044		4044	
1543	4440		4440	
1544	4444		4444	
1545	4044		4044	

/FORWARD GUARD AND REVERSE BLOCK NUMBER

1546	1546	GRZ,	.	
1547	4040		4040	/ON TAPE AS 5145 (OCT)
1550	0440		0440	

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1551 0404      0404

/FORWARD END ZONE

1552 1552      EZM,      ,
1553 0400      0400      /ON TAPE AS 2222 (OCT)
1554 4004      4004
1555 0040      0040

/SUBROUTINE TO SEE IF USER TYPED MARK 384
/TO SPECIFY STANDARD PDP-12 FORMAT
1556 0000      F10PAT, 0
1557 3026      DCA      BLOCKS /CLEAR LOC, BLOCKS IN CASE NOT 10-FORMAT
1560 1031      TAD      TOTAL /AND GET NUMBER TYPED BY USER
1561 1371      TAD      M617 /WAS IT 384?
1562 7640      SZA      CLA
1563 5756      JMP I     F10PAT /NO-RETURN
1564 3020      DCA      W1 /YES-CLEAR W1 FOR WAIT LOOP
1565 1372      TAD      C1101 /AND ADJUST BLOCK TOTAL FOR
1566 3026      DCA      BLOCKS /1102(OCTAL) BLOCKS.
1567 5770      JMP I     ,+1
1570 1306      F10BAK, F10RTN
1571 7161      M617,   -617
1572 1101      C1101,  1101

1573 0001      C1,     0001
1600 1600      *1600

/THE MARK TRACK HAS BEEN WRITTEN, AND TAPE IS
/MOVING FORWARD IN THE FORWARD END ZONE. STOP
/THE TAPE AND SEE IF THERE ARE ANY TAPES LEFT TO
/MARK--IF SO GO DO THEM, ELSE TELL OPERATOR TO THROW THE
/"OFF/WTM" SWITCH TO "OFF"
/HE WILL THEN CONTINUE AFTER THIS ACTION

/KILL WRITE, STOP TAPE

1600 7200      MWTM,   CLA
1601 1027      TAD      DTA /UNIT
1602 6774      SOLC
1603 4777      JMS      NUOTA
1604 5574      JMP I     DOMARK

/MESSAGE TO THE OPERATOR
1605 4502      OFF,    JMS I   TYPE
1606 2305      2305      /SE
1607 2440      2440      /T
1610 2327      2327      /SW
1611 1124      1124      /IT
1612 0310      0310      /CH
1613 4024      4024      /T
1614 1740      1740      /O
1615 1706      1706      /OP
1616 0600      0600      /F
1617 4503      JMS I   TYPIN /WAIT FOR CR
1620 5621      JMP I     ,+1

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1621	3327	SWOFF		/CHECK TO MAKE SURE THAT SWITCH IS OFF
1622	1164	/REVERSE TAPE AND READ MARK TRACK		
1623	1027	PSER, TAD	D13000	/REVERSE GO
1624	6774	TAD	DTA	/UNIT
1625	3020	SDLC		/LOAD COMMAND REGISTER
1626	6773	DCA	W1	/STALL ROUTINE TO GET UP TO SPEED
1627	5226	SDSQ		
1630	6776	JMP	,=1	
1631	2020	SDRC		
1632	5226	ISZ	W1	
1633	6773	JMP	,=4	
1634	7410	SDSQ		/SKIP ON QUAD LINE IF SET AFTER WAIT ROUTINE
1635	5240	SKP		
1636	6771	JMP	,+3	/FLAG WAS SET
1637	5236	SDSS		/READ IN A LINE OF TAPE
1640	6776	JMP	,=1	
1641	6772	SDRC		/READ THE COMMAND REGISTER
1642	7410	SDST		/CHECK FOR A TIMING ERROR
1643	4567	SKP		
1644	0135	JMS	I SELTIM	/TIMING ERROR
1645	1111	AND	MSK77	/CHECK TO SEE IF TAPE IS STILL IN END ZONE
1646	7640	TAD	M55	
1647	5236	SZA	CLA	
1650	4532	JMP	,=11	/NOT A 55 YET
1651	1111	JMS	I SSDSQT	/YES, READ IN SOME MORE
1652	7650	TAD	M55	/IS IT END ZONE
1653	5250	SNA	CLA	
1654	1106	JMP	,=3	/STILL IN END ZONE
1655	1112	TAD	MTR	/GET THE MARK TRACK
1656	7640	TAD	M25	/IS IT EXPAND CODE
1657	4534	SZA	CLA	
1660	7200	JMS	I SCEXPC	/NOT YET, CHECK FOR A 52, AND ADVANCE 3 LINES
1661	1126	CLA		/YES IT IS EXPAND CODE
1662	3127	TAD	M306	/SET UP FOR 198 EXPAND CODES
1663	4532	DCA	CNT	
1664	1112	JMS	I SSDSQT	/THE TAPE SHOULD BE IN SYNC NOW
1665	7640	TAD	M25	/READ THE REST OF EXPAND CODE
1666	4570	SZA	CLA	
1667	2127	JMS	I MARKER	/MARK TRACK ERROR
1670	5263	ISZ	CNT	/INCREMENT COUNTER
1671	1033	JMP	,=5	
1672	3025	TAD	VAR2	/NUMBER OF BLOCKS
1673	4532	DCA	W6	
1674	1112	RSTBLK, JMS	I SSDSQT	/START OF A STANDARD BLOCK
1675	7640	TAD	M25	/FIRST EXPAND CODE AT BEGINNING
1676	4570	SZA	CLA	/OF BLOCK
1677	4532	JMS	I MARKER	/MARK TRACK ERROR
1700	1113	JMS	I SSDSQT	/READ MARK BLOCK NUMBER
1701	7640	TAD	M26	
1702	4570	SZA	CLA	
1703	4532	JMS	I MARKER	/MARK TRACK ERROR
1704	1114	JMS	I SSDSQT	/READ MARK GUARD
1705	7640	TAD	M32	
1706	4570	SZA	CLA	
		JMS	I MARKER	/MARK TRACK ERROR

1727	1130	TAD	M4	
1710	3127	DCA	CNT	
1711	4532	JMS	I SSOSQT	/READ L,CK,F,PF
1712	1115	TAD	M10	
1713	7640	SZA	CLA	
1714	4570	JMS	I MARKER	/MARK TRACK ERROR
1715	2127	ISZ	CNT	
1716	5311	JMP	,=5	
1717	7300	CLA	CLL	
1720	1032	TAD	VAR1	
1721	7004	RAL		
1722	3024	DCA	W5	/NUMBER OF DATA MARKS
1723	4532	JMS	I SSOSQT	/READ DATA MARKS
1724	1116	TAD	M70	
1725	7640	SZA	CLA	
1726	4570	JMS	I MARKER	/MARK TRACK ERROR
1727	2024	ISZ	W5	/COUNT FOR NUMBER OF BLOCKS
1730	5323	JMP	,=5	
1731	1130	TAD	M4	
1732	3127	DCA	CNT	
1733	4532	JMS	I SSOSQT	/READ PF,F,CK,L
1734	1117	TAD	M73	
1735	7640	SZA	CLA	
1736	4570	JMS	I MARKER	/MARK TRACK ERROR
1737	2127	ISZ	CNT	
1740	5333	JMP	,=5	
1741	4532	JMS	I SSOSQT	/READ REVERSE GUARD
1742	1120	TAD	M51	
1743	7640	SZA	CLA	
1744	4570	JMS	I MARKER	
1745	4532	JMS	I SSOSQT	/READ BLOCK NUMBER
1746	1121	TAD	M45	
1747	7640	SZA	CLA	
1750	4570	JMS	I MARKER	/MARK TRACK ERROR
1751	4532	JMS	I SSOSQT	/READ EXPAND CODE
1752	1112	TAD	M25	
1753	7640	SZA	CLA	
1754	4570	JMS	I MARKER	/END OF ONE BLOCK, MARK TRACK ERROR
1755	2025	ISZ	W6	/FINISHED ALL BLOCKS
1756	5273	JMP	RSTBLK	/NO 100 OTHER BLOCKS
1757	1131	TAD	M307	/SET UP FOR INTERBLOCK SYNC AT END OF TAPE
1760	3127	DCA	CNT	
1761	4532	JMS	I SSOSQT	/CHECK FOR 199 EXPAND CODES
1762	1112	TAD	M25	
1763	7640	SZA	CLA	
1764	4570	JMS	I MARKER	/MARK TRACK ERROR
1765	2127	ISZ	CNT	
1766	5361	JMP	,=5	
1767	4532	JMS	I SSOSQT	
1770	1122	TAD	M22	
1771	7640	SZA	CLA	
1772	4570	JMS	I MARKER	
1773	1027	TAD	D1A	

1774	6774		SDLC		
1775	5776		JMP I	,+1	
1776	2000	WDBLKN, DBLKN		/GO OUT TO WRITE DATA AND BLOCK NUMBERS FORWARD	
1777	3133				
	2000	*2000			
2000	1356	DBLKN,	TAD	C2	
2001	3030		DCA	PHASE	
2002	1033		TAD	VAR2	/NUMBER OF BLOCKS
2003	3025		DCA	W6	
2004	3137		DCA	BLK	/INITIAL BLOCK IS 0
2005	1137		TAD	BLK	
2006	4477		JMS I	MESS	/COMPUTE THE COMP OBVERSE OF REV BLK
2007	3140		DCA	REVBK	
2010	6775		SDLD		
2011	1161		TAD	DT1400	/FORWARD,WRITE,GO
2012	1027		TAD	DTA	/UNIT
2013	6774		SDLC		/LOAD THE COMMAND REGISTER
2014	6776		SDRC		/CHECK TO MAKE SURE WRITE IS SET
2015	7006		RTL		
2016	7004		RAL		
2017	7700		SMA	CLA	
2020	4357		JMS	WLO	/WRITE FAILED TO SET
2021	1062		TAD	M6	
2022	3127		DCA	CNT	
2023	6773		SDSQ		/ROUTINE TO GET UP TO SPEED
2024	5223		JMP	,=1	
2025	6775		SDLD		
2026	2127		ISZ	CNT	
2027	5223		JMP	,=4	
2030	6775		SDLD		
2031	6772		SDST		
2032	7410		SKP		
2033	4567		JMS I	SELTIM	/TIMING ERROR
2034	6771	LINE,	SDSS		/WRITE ALL ZEROES TO THE FIRST BLOCK
2035	5234		JMP	,=1	
2036	6775		SDLD		/LOAD THE DATA BUFFER
2037	6776		SDRC		
2040	6772		SDST		
2041	7410		SKP		
2042	4567		JMS I	SELTIM	/TIMING ERROR
2043	0135		AND	MSK77	
2044	3106		DCA	MTR	
2045	1106		TAD	MTR	
2046	1113		TAD	M26	
2047	7640		SZA	CLA	
2050	5234		JMP	LINE	
2051	6775		SDLD		
2052	6772		SDST		
2053	7410		SKP		
2054	4567		JMS I	SELTIM	/TIMING ERROR
2055	5265		JMP	WDBLK	/GO AND WRITE REVERSE GUARD
2056	7300	WDBLK,	CLA CLL		/BEGINNING OF BLOCK,WRITE DATA AND BLOCK NUMBER

2057	4346	JMS	W4L	/WRITE EIGHT LINES
2060	4346	JMS	W4L	/END OF EXPAND CODE, BEGINNING OF BLK NUMBER
2061	1137	TAD	BLK	/GET FORWARD BLOCK NUMBER
2062	4346	JMS	W4L	/WRITE IT
2063	7200	CLA		
2064	4346	JMS	W4L	/WRITE FIRST WORD OF REV GUARD
2065	7200	WDOBLK, CLA		
2066	4346	JMS	W4L	/SECOND WORD OF REVERSE GUARD
2067	4346	JMS	W4L	
2070	4346	JMS	W4L	/FIRST WORD OF REVERSE CHECKSUM
2071	1031	WDATA, TAD	TOTAL	/NUMBER OF DATA WORDS TO BE WRITTEN
2072	7041	CIA		
2073	3024	DCA	W5	/SET UP COUNTER
2074	4346	JMS	W4L	
2075	2024	ISZ	W5	/INCREMENT COUNTER
2076	5274	JMP	.-2	
2077	7300	CLA	CLL	
2100	1135	TAD	MSK77	/COME BACK TO WRITE LAST WORD AND CHECKSUM
2101	4346	JMS	W4L	
2102	7200	CLA		
2103	4346	JMS	W4L	/FINISH CHECKSUM
2104	4346	JMS	W4L	/FIRST WORD OF REVERSE LOCK
2105	4346	JMS	W4L	/LAST WORD OF RL AND HALF OF GUARD
2106	4346	JMS	W4L	/REST OF GUARD
2107	1140	TAD	REVBK	/GET REVERSE BLOCK NUMBER
2110	4346	JMS	W4L	
2111	7240	CLA	CMA	
2112	4346	JMS	W4L	/END OF BLOCK NUMBER AND HALF OF EXPAND CODE
2113	4346	JMS	W4L	/END OF EXPAND CODE
2114	2137	ISZ	BLK	
2115	7200	CLA		
2116	1137	TAD	BLK	
2117	4477	JMS	I MESS	/COMPUTE NEW BLK NUMBER
2120	3140	DCA	REVBK	
2121	6772	SDST		
2122	7410	SKP		
2123	4567	JMS	I SELTIM	/TIMING ERROR
2124	2025	ISZ	W6	/IS IT DONE WRITING BLK AND DATA
2125	5256	JMP	WDOBLK	/NO
2126	6773	SDSQ		
2127	5326	JMP	.-1	
2130	6777	SDRD		
2131	7200	CLA		
2132	1165	TAD	DT1000	/SEARCH FOR END ZONE
2133	1027	TAD	DTA	/GET UNIT
2134	6774	SDLC		/LOAD THE COMMAND REG
2135	6771	SDSS		
2136	5335	JMP	.-1	
2137	6776	SDRC		
2140	0135	AND	MSK77	
2141	1122	TAD	M22	
2142	7640	SZA	CLA	
2143	5335	JMP	.-6	
2144	5745	JMP	I .+1	
2145	2400	DBLOCK		

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2146 0000      W4L,      0
2147 6773      SDSQ
2150 5347      JMP      .-1      /SKIP ON QUAD LINE FLAG
2151 6775      SDLD      /LOAD THE DATA BUFFER
2152 6772      SDST      /CHECK FOR A TIMING ERROR
2153 7410      SKP
2154 4567      JMS      I SELTIM      /TIMING ERROR
2155 5746      JMP      I W4L

2156 0002      C2,      0002

2157 0000      WLO,      0
2160 1027      TAD      DTA      /STOP THE TAPE
2161 6774      SDLC      /LOAD THE COMMAND REGISTER
2162 4502      JMS      I TYPE
2163 2003      2003      /PC
2164 4000      4000      /END
2165 7240      CLA      CMA
2166 1357      TAD      WLO
2167 4501      JMS      I TYOCT
2170 4502      JMS      I TYPE
2171 4040      4040
2172 2722      2722      /WR
2173 1124      1124      /IT
2174 0540      0540      /E
2175 0000      0000      /END
2176 5777      JMP      I .+1
2177 0744      ZCOM

2200 2200      *2200
2200 1372      BLCS0,    TAD      C4
2201 3030      DCA      PHASE
2202 7300      CLA      CLL
2203 1033      TAD      VAR2
2204 3025      DCA      W6      /SET UP FOR THE NUMBER OF BLOCKS
2205 3137      DCA      BLK      /SET BLK TO 0
2206 1165      TAD      DT1000  /FORWARD READ
2207 1027      TAD      DTA      /UNIT
2210 6774      SDLC      /LOAD THE COMMAND REG
2211 1137      TAD      BLK
2212 4477      JMS      I MESS      /CALCULATE THE COMPLEMENT OBVERSE
2213 3140      DCA      REVBK
2214 6772      SDST
2215 7410      SKP
2216 4567      JMS      I SELTIM      /TIMING ERROR
2217 1062      TAD      M6      /WAIT TO GET UP TO SPEED
2220 3127      DCA      CNT      /SET UP COUNTER
2221 6773      SDSQ      /SKIP ON A QUAD LINE FLAG
2222 5221      JMP      .-1
2223 6777      SDRD      /READ THE DATA BUFFER TO CLEAR FLAG
2224 2127      ISZ      CNT
2225 5221      JMP      .-4

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2226	7200	CLA		
2227	3142	BLCSDA, DCA	CHKSUM	
2230	4507	JMS I	SLRDRC	/READ A SINGLE LINE AT A TIME
2231	1113	TAD	M26	
2232	7640	SZA	CLA	/IS IT BLOCK MARK
2233	5777	JMP	SRDRC+4	/NO, GO BACK
2234	6772	SDST		
2235	7410	SKP		
2236	4567	JMS I	SELTIM	/TIMING ERROR
2237	1110	TAD	DATRD	
2240	7041	CIA		
2241	1137	TAD	BLK	
2242	7640	SZA	CLA	
2243	4571	JMS I	BLKERR	/BLK NUMBER ERROR
2244	4532	JMS I	SSDSQT	/READ GUARD
2245	4532	JMS I	SSDSQT	/READ REVERSE LOCK
2246	4532	JMS I	SSDSQT	/READ CHECKSUM
2247	6777	SDRD		/READ THE DATA BUFFER
2250	6772	SDST		
2251	7410	SKP		
2252	4567	JMS I	SELTIM	/TIMING ERROR
2253	0135	AND	MSK77	
2254	4541	JMS I	BCXOR	/GO OUT TO CHECKSUM ROUTINE
2255	1031	RDATA, TAD	TOTAL	/NUMBER OF WORDS PER BLOCK
2256	7041	CIA		
2257	3024	DCA	W5	/SET UP COUNTER
2260	6773	SDSQ		
2261	5260	JMP	,=1	
2262	6777	SDRD		/READ THE DATA BUFFER
2263	6772	SDST		
2264	7410	SKP		
2265	4567	JMS I	SELTIM	/TIMING ERROR
2266	3110	DCA	DATRD	
2267	1110	TAD	DATRD	/SAVE THE DATA WORD
2270	7640	SZA	CLA	
2271	4572	JMS I	DATERR	/DATA ERROR
2272	1110	TAD	DATRD	
2273	4541	JMS I	BCXOR	
2274	6772	SDST		/CHECK FOR A TIMING ERROR
2275	7410	SKP		
2276	4567	JMS I	SELTIM	/TIMING ERROR
2277	2024	ISZ	W5	
2300	5260	JMP	RDATA+3	
2301	6773	SDSQ		/READ REVERSE CHECKSUM
2302	5301	JMP	,=1	
2303	6777	SDRD		/READ IT IN
2304	6772	SDST		
2305	7410	SKP		
2306	4567	JMS I	SELTIM	/TIMING ERROR
2307	0052	AND	C7700	
2310	4541	JMS I	BCXOR	/CHECK CHECK SUM
2311	1142	TAD	CHKSUM	
2312	0135	AND	MSK77	
2313	7001	IAC		
2314	1052	TAD	C7700	

2315	7640	SZA	CLA	
2316	4573	JMS	I	CHKERR /CHECKSUM ERROR
2317	6772	SDST		
2320	7410	SKP		
2321	4567	JMS	I	SELTIM /TIMING ERROR
2322	4537	JMS	I	SLRDR /ADVANCE A SINGLE LINE FLAG
2323	1125	TAD	M31	/LOOK FOR REV BLK NUMBER
2324	7640	SZA	CLA	
2325	5777	JMP		SRDR+4
2326	6772	SDST		
2327	7410	SKP		
2330	4567	JMS	I	SELTIM /TIMING ERROR
2331	1110	TAD	DATRO	
2332	7041	CIA		
2333	1140	TAD	REVBK	/COMPARE BLOCK READ WITH ONE COMPUTED
2334	7640	SZA	CLA	
2335	4571	JMS	I	BLKERR /BLOCK NUMBER ERROR
2336	6773	SDSQ		
2337	5336	JMP		.-1
2340	6777	SDRD		
2341	6772	SDST		
2342	7410	SKP		
2343	4567	JMS	I	SELTIM /TIMING ERROR
2344	7300	CLA	CLL	
2345	2137	ISZ	BLK	
2346	1137	TAD	BLK	
2347	4477	JMS	I	MESS
2350	3140	DCA	REVBK	
2351	6772	SDST		
2352	7410	SKP		
2353	4567	JMS	I	SELTIM /TIMING ERROR
2354	2025	ISZ	W6	
2355	5227	JMP		BLCSDA
2356	1165	TAD	D1000	
2357	1027	TAD	D1A	
2360	6774	SDLC		
2361	6771	SDSS		
2362	5361	JMP		.-1
2363	6776	SDRC		
2364	0135	AND	MSK77	
2365	1122	TAD	M22	
2366	7640	SZA	CLA	
2367	5361	JMP		.-6
2370	5771	JMP	I	.+1
2371	2442	RDBLKS		
2372	0004	C4,	0004	
2377	3124			
	2400	*2400		
2400	1240	DBLOCK,	TAD	C3
2401	3030		DCA	PHASE
2402	7300		CLA	CLL
2403	3235		DCA	DISBLK
2404	1164		TAD	D13000 /REVERSE,GO

2405	1027	TAD	DTA	/UNIT
2406	6774	SDLC		/LOAD THE COMMAND REGISTER
2407	7300	CLA	CLL	
2410	6771	DISLUP,	SDSS	
2411	5210	JMP	,=1	
2412	7300	CLA	CLL	
2413	6777	SDRD		
2414	3236	DCA	DISDAT	/SAVE THE DATA BUFFER
2415	6776	SDRC		
2416	0135	AND	MSK77	/MASK OUT THE MARK TRACK
2417	1113	TAD	M26	/CHECK FOR BLOCK NUMBER
2420	7440	SZA		
2421	5226	JMP	DISEND	/NOT BLK MARK,CHECK FOR END ZONE
2422	1236	TAD	DISDAT	/DISPLAY THE NUMBER IN THE AC
2423	2235	ISE	DISBLK	
2424	5223	JMP	,=1	
2425	5210	JMP	DISLUP	/GO SEARCH FOR THE NEXT BLOCK
2426	1237	DISEND,	TAD	FOUR
2427	7640	SZA	CLA	/IS IT END ZONE
2430	5210	JMP	DISLUP	/NO,GO GET NEXT LINE
2431	1027	TAD	DTA	/STOP GET READY TO READ
2432	6774	SDLC		/LOAD THE COMMAND REGISTER
2433	5634	JMP	,=1	
2434	2200	BLCSO		
2435	0000	DISBLK,	0	
2436	0000	DISDAT,	0	
2437	0004	FOUR,	4	
2440	0003	C3,	0003	
2441	0005	C5,	0005	
2442	1241	RDBLKS,	TAD	C5
2443	3030	DCA	PHASE	
2444	1033	TAD	VAR2	
2445	3024	DCA	W5	/SET UP FOR NUMBER OF BLOCKS
2446	7001	IAC		
2447	1033	TAD	VAR2	
2450	3025	DCA	W6	/SET UP TO CHECK BLK REVERSE
2451	1164	TAD	D13000	/READ REVERSE GO
2452	1027	TAD	DTA	/UNIT
2453	6774	SDLC		/LOAD THE COMMAND REGISTER
2454	1062	TAD	M6	
2455	3127	DCA	CNT	
2456	6771	SDSS		
2457	5256	JMP	,=1	
2460	6776	SDRC		
2461	7200	CLA		
2462	2127	ISE	CNT	
2463	5256	JMP	,=5	
2464	6771	RDBLK,	SDSS	
2465	5264	JMP	,=1	
2466	6777	SDRD		/READ THE DATA BUFFER AND STORE IT AWAY
2467	3127	DCA	CNT	
2470	6776	SDRC		
2471	0135	AND	MSK77	
2472	1113	TAD	M26	

2473	7640	SZA	CLA	/IS IT BLOCK NUMBER
2474	5264	JMP	RDBLK	
2475	1127	TAD	CNT	
2476	1025	TAD	W6	
2477	7640	SZA	CLA	
2500	4571	JMS	I BLKERR	/BLOCK NUMBER ERROR
2501	7001	IAC		
2502	1025	TAD	W6	/INCREMENT A NUMBER FOR COMPARE COUNTER
2503	3025	DCA	W6	
2504	2024	ISZ	W5	/INCREMENT BLK COUNTER
2505	5264	JMP	RDBLK	
2506	6771	SDSS		
2507	5306	JMP	,=1	
2510	6776	SDRC		
2511	0135	AND	MSK77	
2512	1122	TAD	M22	
2513	7640	SZA	CLA	
2514	5306	JMP	,=6	
2515	1027	TAD	DTA	
2516	6774	SDLC		/LOAD THE COMMAND REGISTER WITH UNIT STOP
2517	7001	IAC		
2520	3030	DCA	PHASE	
2521	4777	JMS	NUDTA	
2522	5776	JMP	PSER	
2523	5724	JMP	I ,=1	
2524	1061	INIT		/END GO BACK TO DIRECT
2576	1622			
2577	3133			
	2600			
		*2600		
2600	7300	RDFA,	CLA	CLL
2601	1164		TAD	DT3000
2602	1027		TAD	DTA
2603	6774		SDLC	/REVERSE READ GO
2604	6771		SDSS	/GET UNIT
2605	5204		JMP	/LOAD THE COMMAND REGISTER
2606	6776		JMP	/SKIP ON A SINGLE LINE FLAG
2607	0135		SDRC	
2610	1122		AND	/READ THE COMMAND REGISTER
2611	7640		TAD	MSK77
2612	5204		SZA	M22
2613	1165		JMP	/IS IT END ZONE
2614	1027		TAD	CLA
2615	6774		SDLC	/YES
2616	1062		TAD	/NO GO BACK AND LOOK AGAIN
2617	3127		DCA	/FORWARD READ GO
2620	6771		SDSS	/UNIT
2621	5220		JMP	/LOAD THE COMMAND REGISTER
2622	6776		SDRC	
2623	7200		CLA	
2624	2127		ISZ	
2625	5220		JMP	CNT
2626	1113	RDFAI,	TAD	,=5
2627	3022		DCA	M26
				W3
				/SET UP COUNTER TO READ 22 BLOCKS

```

2630 1267      TAD      BADD      /SET UP BUFFER ADDRESS
2631 3011      DCA      X2
2632 6771      SDSS      /GO SINGLE LINE FLAGS
2633 5232      JMP      .-1
2634 6777      SDRD      /READ THE DATA BUFFER
2635 3127      DCA      CNT
2636 6776      SDRD      /READ THE COMMAND REGISTER
2637 2135      AND      MSK77
2640 1113      TAD      M26      /SEARCH FOR BLOCK NUMBER
2641 7640      SZA      CLA
2642 5232      JMP      RDA1+4 /NOT BLOCK NUMBER YET GO BACK AGAIN
2643 1127      TAD      CNT      /OK BLK NUMBER STORE IT AWAY
2644 3411      DCA      X2
2645 2022      ISZ      W3      /INCREMENT COUNTER
2646 5232      JMP      RDA1+4 /NOT 22 BLOCKS YET
2647 1027      TAD      DTA
2650 6774      SOLC      /STOP THE DTA

```

/TYPE OUT BLOCK NUMBERS AND DTA UNIT#

```

2651 4502      JMS      I TYPE
2652 0424      0424      /DT
2653 0140      0140      /A
2654 0000      0000      /END
2655 1027      TAD      DTA      /GET UNIT NUMBER
2656 7006      RTL
2657 4501      JMS      I TYOCT /AND TYPE IT OUT
2660 4502      JMS      I TYPE
2661 4345      4345      /CR&LF
2662 0000      0000      /END
2663 1113      TAD      M26      /WILL TYPE ALL
2664 3020      DCA      W1      /22 WORDS
2665 1067      TAD      BADD      /ADDRESS OF BLOCK
2666 3011      DCA      X2      /NUMBERS TO INDEX
2667 1411      TAD      I X2      /FIRST OR NEXT BLOCK
2670 4501      JMS      I TYOCT /TYPE IT OUT
2671 4502      JMS      I TYPE /CR&LF
2672 4345      4345      /CR&LF
2673 0000      0000      /END
2674 2020      ISZ      W1      /COMPLETE
2675 5267      JMP      .-6
2676 5472      JMP      I IT      /GO ASK FOR FORMAT

2677 7300      RDR,      CLA      CLL
2700 1165      TAD      DT1000 /FORWARD READ GO
2701 1027      TAD      DTA      /UNIT
2702 6774      SOLC      /LOAD THE COMMAND REGISTER
2703 6771      SDSS      /SKIP ON A SINGLE LINE FLAG
2704 5303      JMP      .-1
2705 6776      SDRD      /READ THE COMMAND REGISTER
2706 0135      AND      MSK77
2707 1122      TAD      M22      /CHECK FOR END ZONE
2710 7640      SZA      CLA
2711 5303      JMP      .-6      /NOT YET GO BACK
2712 1164      TAD      DT3000 /REVERSE READ GO

```

2713	1027	TAD	DTA	/UNIT
2714	6774	SDLC		/LOAD THE COMMAND REGISTER
2715	1062	TAD	M6	
2716	3127	DCA	CNT	
2717	6771	SDSS		
2720	5317	JMP	.-1	
2721	6776	SDRC		
2722	7200	CLA		
2723	2127	ISE	CNT	
2724	5317	JMP	.-5	
2725	5226	JMP	RDFa1	/STORE NUMBERS IN REVERSE
2726	4503	RETRY, JMS	I TYPIN	
2727	4471	JMS	I COMPAR	
2730	0322			/R
2731	0305			/E
2732	0324			/T
2733	0322			/R
2734	0331			/Y
2735	0000			/END
2736	5472	JMP	I IT	/GUESS HE DOESN'T WANT TO TRY AGAIN
2737	7200	CLA		
2740	1165	TAD	Df1000	/FORWARD READ GO
2741	1027	TAD	DfA	/UNIT
2742	6774	SDLC		/LOAD THE COMMAND REGISTER
2743	1062	TAD	M6	
2744	3127	DCA	CNT	/WAIT 6 LINES
2745	6771	SDSS		
2746	5345	JMP	.-1	
2747	6776	SDRC		/READ THE COMMAND REGISTER
2750	2127	ISE	CNT	
2751	5345	JMP	.-4	
2752	6771	SDSS		
2753	5352	JMP	.-1	
2754	6776	SDRC		
2755	0135	AND	MSK77	
2756	1122	TAD	M22	
2757	7640	SZA	CLA	
2760	5352	JMP	.-6	
2761	1164	TAD	Df3000	
2762	1027	TAD	DfA	
2763	6774	SDLC		
2764	7201	CLA	IAC	
2765	3030	DCA	PHASE	
2766	5767	JMP	I .-1	
2767	1633	PSER+11		
3000				
3000	0000	SDSQf,	0	
3001	6773	SDSQ		/ADVANCE SIX LINES
3002	5201	JMP	.-1	/SKIP ON QUAD LINE FLAG
3003	6776	SDRC		/READ COMMAND REGISTER

3024	6772	SDST		
3025	7410	SKP		
3026	4567	JMS I	SELTIM	/TIMING ERROR
3027	6771	SDSS		
3028	5227	JMP	,=1	/SKIP ON SINGLE LINE FLAG
3029	6776	SDRC		
3030	6772	SDST		
3031	7410	SKP		
3032	4567	JMS I	SELTIM	/TIMING ERROR
3033	6771	SDSS		
3034	5215	JMP	,=1	
3035	6776	SDRC		/READ THE COMMAND REGISTER
3036	6772	SDST		
3037	7410	SKP		
3038	4567	JMS I	SELTIM	/TIMING ERROR
3039	0135	AND	MSK77	/SAVE THE MARK TRACK LAST 6 BITS
3040	3106	DCA	MTR	
3041	1106	TAD	MTR	
3042	5600	JMP I	SDSQ	
3027	0000	A3LNS,	0	/ADVANCE THREE LINES
3030	6771	SDSS		
3031	5230	JMP	,=1	/SKIP ON SINGLE LINE FLAG
3032	6776	SDRC		
3033	6772	SDST		
3034	7410	SKP		
3035	4567	JMS I	SELTIM	/TIMING ERROR
3036	6771	SDSS		
3037	5236	JMP	,=1	
3040	6776	SDRC		
3041	6772	SDST		
3042	7410	SKP		
3043	4567	JMS I	SELTIM	/TIMING ERROR
3044	6771	SDSS		
3045	5244	JMP	,=1	
3046	6776	SDRC		
3047	6772	SDST		
3050	7410	SKP		
3051	4567	JMS I	SELTIM	/TIMING ERROR
3052	0135	AND	MSK77	
3053	3106	DCA	MTR	
3054	1106	TAD	MTR	
3055	5627	JMP I	A3LNS	
3056	0000	CEXPC,	0	
3057	1106	TAD	MTR	
3060	1124	TAD	M52	
3061	7640	SZA	CLA	
3062	4570	JMS I	MARKER	/MARK TRACK ERROR
3063	4227	JMS	A3LNS	/READ THREE MORE LINES
3064	1112	TAD	M25	/IS IT 25 NOW
3065	7640	SZA	CLA	
3066	4570	JMS I	MARKER	/NO ,MARK TRACK ERROR
3067	5656	JMP I	CEXPC	/YES!! IT IS EXPAND CODE NUMBER I

/SIXBIT COMPLEMENT XOR SUBROUTINE
 /SUBROUTINE IS ENTERED WITH DATA WORD TO BE XORED IN AC
 /TWO SIX-BIT COMPLEMENT XORS WILL TAKE PLACE TO LOC CHKSUM
 /WITH THE RESULT IN CHKSUM

```

3070 2000 SBCXOR, 0
3071 7040 CMA /COMPLEMENT WORD
3072 3143 DCA SBWORD /AND SAV
3073 1143 TAD SBWORD
3074 0142 AND CHKSUM
3075 7041 CIA
3076 7104 CLL RAL
3077 1143 TAD SBWORD
3100 1142 TAD CHKSUM
3101 3142 DCA CHKSUM
3102 1143 TAD SBWORD
3103 7112 RTR CLLIRTRIRTR
3104 7012
3105 7012
3106 3143 DCA SBWORD
3107 1143 TAD SBWORD
3110 0142 AND CHKSUM
3111 7041 CIA
3112 7104 CLL RAL
3113 1143 TAD SBWORD
3114 1142 TAD CHKSUM
3115 0135 AND MSK77
3116 3142 DCA CHKSUM
3117 5670 JMP I SBCXOR

3120 0000 SRDRC, 0
3121 6773 SDSQ
3122 7410 SKP
3123 5326 JMP ,+3
3124 6771 SDSS
3125 5324 JMP ,=1
3126 6777 SDRD
3127 3110 DCA DATRD
3130 6776 SDRD
3131 0135 AND MSK77
3132 5720 JMP I SRDRC

3133 0000 NUDTA, 0
3134 1754 TAD I LSTPT /GET CURRENT VALUE OF DATA LIST PTR
3135 3353 DCA TBUFPT /STORE IT AS TEM,BUF,PTR
3136 1753 TAD I TBUFPT /GET A DTA # FROM THE LIST
3137 0037 AND C0007
3140 7640 SZA CLA /IS IT A 7777
3141 5346 JMP LSTEND /YES END OF LIST
3142 1753 TAD I TBUFPT /NO;GET IT BACK
3143 3027 DCA DIA
3144 2754 ISZ I LSTPT /INCREMENT LIST POINTER
3145 5733 JMP I NUDTA /RETURN
/COME HERE AT END OF LIST TO RESET POINTERS AND RETURN TO CALL+2
3146 2333 LSTEND, ISZ NUDTA /INCREMENT RETURN POINTER
  
```

```

3147 1752      TAD I STRPT /GET ADR OF START OF LIST
3150 3754      DCA I LSTPT
3151 5334      JMP NUDTA+1 /GO GET FIRST DTA# AND RETURN
3152 1176      STRPT, DBUFAD /POINTER TO START OF DATA LIST
3153 2000      TBUFPT, 0 /TEM STORAGE FOR BOT PTR
3154 1175      LSTPT, DBUFPT /POINTER TO CURRENT VALUE OF DTA LIST PTR

/CONSTANTS FOR FORMULA TRANSLATION SECTION
3155 3156      BINCON, ,+1
3156 0001      0001
3157 0012      0012
3160 0144      0144
3161 1750      1750
3162 0000      DTABUF, 0

3200 *3200
3200 4502      SWCHK, /CHECK SWITCH TO SEE IF SET TO WTM POSITION
3201 2305      JMS I TYPE /TYPE OUT MESSAGE
3202 2440      2305 /SE
3203 2327      2440 /T
3204 1124      2327 /SW
3205 0310      1124 /IT
3206 4024      0310 /CH
3207 1740      4024 /T
3210 2724      1740 /O
3211 1500      2724 /WT
3212 4503      1500 /M
3213 7200      JMS I TYPIN /WAIT FOR CR
3214 3256      CLA
3215 6775      DCA CNTRL
3216 6771      SDLD /CLEAR SINGLE AND QUAD FLAGS
3217 7410      SDSS
3220 5224      SKP
3221 2256      JMP ,+4
3222 5216      ISZ CNTRL
3223 5267      JMP ,+4
3224 6774      RSTSH, /SEE IF THE DRIVE IS OK
3225 6775      SDLC /LOAD CR TO CLEAR TIMEING ERROR
3226 1162      SDLD /LOAD DATA BUFFER TO CLEAR S Q FLAGS
3227 1027      TAD DT0400 /SET WRITE
3230 3257      TAD DTA /GET UNIT
3231 1257      DCA SAV /STORE IT AWAY
3232 6771      TAD SAV
3233 5232      SDSS
3234 6774      JMP ,+1
3235 1257      SDLC
3236 6774      TAD SAV
3237 6776      SDRC /LOAD THE TRANSPORT
3240 7006      RTL /READ THE COMMAND REGISTER AND CHECK IT
3241 7004      RAL

```

3242	7520	SMA		/CHECK WRITE TO BE SET
3243	5260	JMP	ERCHK	/WRITE IS NOT SET
3244	7024	RAL		/CHECK WLO
3245	7510	SPA		
3246	5260	JMP	ERCHK	/WLO
3247	7004	RAL		/CHECK SELECT AND TIMING ERROR
3250	7710	SPA	CLA	
3251	5260	JMP	ERCHK	/SELECT OR TIMING ERROR
3252	4777	JMS	NUDTA	/CHECK OTHER DRIVE IF ANY
3253	5213	JMP	RSTSM=11	/CHECK OTHER DRIVE
3254	5655	JMP	I ,+1	
3255	1400	STMK		
3256	2000	CNTERL,	0	
3257	2000	SAV,	0	
3260	4502	ERCHK,	JMS I TYPE	/INCORRECT SETUP
3261	2305		2305	/SE
3262	2425		2425	/TU
3263	2077		2077	/P
3264	0000		0000	/END
3265	5666		JMP I ,+1	
3266	1000		START	
3267	4502	SWCHER,	JMS I TYPE	
3270	2327		2327	/SW
3271	1124		1124	/IT
3272	0310		0310	/CH
3273	4016		4016	/N
3274	1724		1724	/OT
3275	4023		4023	/S
3276	0524		0524	/ET
3277	4024		4024	/T
3300	1740		1740	/O
3301	2724		2724	/WT
3302	1540		1540	/M
3303	1722		1722	/OR
3304	4023		4023	/S
3305	1116		1116	/IN
3306	0714		0714	/GL
3307	0540		0540	/E
3310	1411		1411	/LI
3311	1605		1605	/NE
3312	4006		4006	/F
3313	1401		1401	/LA
3314	0740		0740	/G
3315	0601		0601	/FA
3316	1114		1114	/IL
3317	0504		0504	/ED
3320	4024		4024	/T
3321	1740		1740	/O
3322	2305		2305	/SE
3323	2440		2440	/T
3324	4543		4543	/CR LF
3325	0000		0000	/END
3326	5200	JMP	SWCHK	

```

3327 7200 SWOFF, CLA
3330 3256 DCA CNTERL
3331 6775 SOLD /CLEAR ANY FLAGS THAT ARE SET
3332 6771 SDSS
3333 7410 SKP
3334 5776 JMP OFF /FLAG SHOULDN'T BE SET
3335 2256 ISZ CNTERL
3336 5332 JMP ,+4
3337 7200 CLA
3340 5741 JMP I ,+1
3341 1622 PSER

```

```

3376 1605
3377 3133
      3400 *3400
      /INPUT BUFFER FOR TELETYPE THIS MUST BE AT THE END OF PROGRAM
3400 0000 BUFFER, 0
$

```


4000
4100

4200
4300

4400
4500

4600
4700

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

A3LNS	3027	CRCOD	0055	M14	0064	QUX	0447
BADD	0067	CRFLAG	1146	M143	0123	RDATA	2255
BCXOR	0141	DATERR	0172	M144	0065	RDBLK	2464
BFR	0070	DATRD	0110	M2	0060	RDBLKS	2442
BINCO	0166	DBLKN	2000	M22	0122	RDFA	2630
BINCON	3155	DBLOCK	2400	M25	0112	RDFA1	2626
BLCSO	2220	DBUFAD	1176	M26	0113	RDR	2677
BLCSDA	2227	DBUFPT	1175	M3	0061	REPEAT	1147
BLK	0137	DCTR	1174	M306	0126	RETRY	2726
BLKERR	0171	DISBLK	2435	M307	0131	REVLK	0140
BLOCKS	0026	DISDAT	2436	M31	0125	REZ	1510
BUFFER	3400	DISEND	2426	M32	0114	RSEND	0144
C0007	0037	DISLUP	2410	M4	0130	RSTBLK	1673
C0017	0034	DIV3	1244	M40	0253	RSTM	3224
C0070	0035	DNC	1203	M45	0121	SA3LNS	0133
C0077	0036	DNUM	1177	M51	0120	SAV	3257
C0700	0040	DDCOMP	1164	M52	0124	SBCXOR	3070
C1	1573	DOMARK	0174	M55	0111	SBWORD	0143
C1101	1572	DT0400	0162	M6	0062	SCEXP	0134
C1620	0050	DT1000	0165	M617	1571	SOLC	6774
C2	2156	DT1400	0161	M7	0063	SOLD	6775
C201	0042	DT2000	0163	M70	0116	SDRC	6776
C203	0041	DT3000	0164	M73	0117	SDRD	6777
C212	0255	DTA	0027	MARK	1200	SOSQ	6773
C215	0256	DTABUF	3162	MARKER	0170	SOSQT	3000
C245	0257	DTRK	1434	MES	0454	SDSS	6771
C260	0043	DZ	1533	MESSAGE	0202	SDST	6772
C261	0044	ERCHK	3260	MESS	0077	SELTIM	0167
C270	0045	E2M	1552	MSK77	0135	SETUP	1470
C271	0046	F10BAK	1570	MSRGHT	0221	SETUP1	1500
C277	0047	F10PAT	1556	MTP	0246	SLRDRC	0107
C3	2440	F10RTN	1306	MTR	0106	SPCOD	0066
C340	0254	FBM	1520	MWTM	1600	SRDRC	3120
C4	2372	FEZ	1537	NTYRTN	0265	SSOSQT	0132
C5	2441	FORM10	1355	NUD	0136	STALL	0600
C6060	0377	FOUR	2437	NUDTA	3133	START	1000
C7000	0051	GETDTA	1145	OFF	1605	STHK	1400
C7700	0052	GRB	1442	OKCR	1044	STRTP	3152
C7714	0053	GRZ	1546	PATCH	0515	STX	0100
C7761	0054	IBS	1242	PFCRC	1440	SWCHER	3267
CERR	0332	IBZ	1514	PFORM	0763	SWCHK	3200
CEXPC	3056	INBLSY	1422	PHASE	0030	SWOFF	3327
CEZ	1410	INIT	1061	PSER	1622	TBUFFT	3153
CHKERR	0173	INIT1	1055	Q1	0400	TFORM	0756
CHKSUM	0142	IT	0072	Q2	0410	TOTAL	0031
CNT	0127	LETK	0056	Q3	0422	TW1	0376
CNTERL	3256	LFCOD	0057	Q4	0434	TYCT	0336
COMCHK	1156	LINE	2034	QU	0152	TYCT1	0371
COMCTR	1173	LRCFP	1432	QU1	0073	TYOCT	0101
COMPAR	0071	LSTEND	3146	QU2	0074	TYPE	0102
COMPPE	0312	LSTPT	3154	QU3	0075	TYPECH	0222
CRCHK	1016	M10	0115	QU4	0076	TYPIN	0103

TYPN	0260
TYJU	1003
V1	2506
V2	2507
V3	2513
VALCHK	1024
VAR1	2032
VAR2	0033
W1	2020
W2	0021
W3	2022
W4	0023
W4L	2146
W5	0024
W6	0025
WAIT	0104
WC	0105
WDATA	2071
WDBLK	2056
WDBLKN	1776
WDBLK	2065
W02	1430
WEZF	1455
WLMRF	1524
WLO	2157
X1	0010
X2	0011
ZBLK	0613
ZCOM	0744
ZDATA	0637
ZHKT	0657
ZPAR	0702
ZTIM	0724

ERRORS DETECTED: 0

LINKS GENERATED: 7

RUN-TIME: 25 SECONDS

3K CORE USED